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Master Thesis

Pre-trained Deep Learning Networks for Analyzing Hyperspectral Image Data

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Abstract

Hyperspectral imaging has emerged as a powerful tool for various applications including agricultural applications, particularly in crop quality assessment and variety classification. However, the deployment of deep learning models for hyperspectral data analysis faces significant challenges due to limited labeled datasets and the high dimensionality of spectral information. Self-supervised pre-training methods have proven effective in learning robust feature representations from unlabeled data, yet their application to hyperspectral imaging remains underexplored. This thesis introduces a novel Masked Spatial-Spectral Transformer (MaskedSST) architecture that integrates blockwise 3D patch embedding, factorized attention mechanisms, and self-supervised pre-training strategies to address these challenges. The model employs two distinct pre-training objectives: Masked Autoencoding (MAE) with 85% random token masking and Contrastive Learning (SimCLR) using data augmentation. The architecture is built on a factorized transformer encoder with 4 layers, 8 attention heads, and 128-dimensional embeddings, comprising 1,652,609 trainable parameters. Pre-training was conducted on a diverse multi-domain dataset of 4,512 hyperspectral patches spanning six application domains (coffee, apple leaves, paper, sugar compounds, archaeological artifacts, and grapes) to learn transferable cross-domain representations. The model was implemented on a high-performance computing cluster using PyTorch with NVIDIA V100 GPUs. Evaluation on downstream classification task demonstrated that MAE pre-training achieved 88.00% accuracy after fine-tuning, outperforming training from scratch by 3.00%. The factorized spatial-spectral attention reduces attention complexity by approximately $256\times$ compared with full self-attention for the 4×4 configuration. The factorized attention mechanism enabled efficient processing of 4,096 tokens per sample without requiring aggressive downsampling. Furthermore, data volume analysis revealed early pre-training saturation, with initial corpus fractions (20%–40%) yielding the primary accuracy jump and the final 20% adding only approximately 0.1 percentage points in overall accuracy. Investigating multi-domain pre-training showed that adding specific datasets directly shapes downstream performance on the Houston target. While relevant domains like sugar triggered sharp accuracy spikes, unrelated domains temporarily flat-tened progress, highlighting how cross-domain distribution shifts impact generalizability. Limitations include the requirement for substantial pre-training computational resources and sensitivity to spectral calibration quality across heterogeneous data sources. The model demonstrates that self-supervised pre-training on diverse hyperspectral data yields transferable representations. Future enhancements include exploring cross-domain adaptation techniques, investigating optimal masking strategies for spectral continuity, and extending evaluation to additional crops and imaging modalities for broader applicability in precision agriculture.

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Magdeburg, Germany

Gowtham Premkumar

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List of Abbreviations

AVIRIS	Airborne Visible/Infrared Imaging Spectrometer
EnMAP	Environmental Mapping and Analysis Program
GPU	Graphics Processing Unit
HSI	Hyperspectral Imaging
MAE	Masked Autoencoding
MaskedSST	Masked Spatial-Spectral Transformer
MSE	Mean Squared Error
NIR	Near-Infrared
PRISMA	Preadressore Iperspettrale della Missione Applicativa
SimCLR	Simple Framework for Contrastive Learning
SOTA	State-of-the-Art
SWIR	Short-Wave Infrared
VIS	Visible Spectrum
ViT	Vision Transformer

Appendix: STATEMENT OF AUTHORSHIP of the Student

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1

Introduction

This chapter provides an overview of the core motivation behind our work, introducing 3D Spatial-Spectral Vision Transformers and self-supervised pre-training strategies for high-dimensional hyperspectral image analysis. Furthermore, we examine current challenges in hyperspectral learning, define our core technical contributions, and present the primary research questions driving this study. Finally, we conclude with an outline of the overall thesis structure. The remainder of this chapter is organized as follows:

- **Section 1.1** details the motivation for this research, focusing on spatial-spectral feature extraction and self-supervised pre-training paradigms.
- **Section 1.2** defines the formal research questions addressed throughout this work.
- **Section 1.3** highlights our major research and contributions.
- **Section 1.4** provides a concise overview of the structure of the remaining chapters.

1.1 Motivation

Hyperspectral images have recently gained popularity due to its usage in precision agriculture [1], enabling detailed material characterization through the acquisition of spectral information across hundreds of wavelengths bands. Machine learning methods like deep neural networks have become the standard approach for any task using hyperspectral images [25]. However, current practice suffers from a critical limitation, for each new classification task, training must start from scratch. Every new application—whether classifying a different crop type, identifying new material types, or analyzing data from a different sensor—requires collecting a completely new dataset and training an entirely new model from random weight initialization. This approach demands substantial labeled training data and computational resources for every task.

Training deep learning models from scratch for each task requires substantial computational and data. Pretraining a vision transformer [8] for example for 100-200 epochs takes 6 - 10 hr on multi GPU cluster. Repeating this process for every new task costs linearly.

The core concept: Instead of training separate models from scratch we can train a general purpose hyperspectral feature extractor once on a large collection of unlabelled hyperspectral images. This Pre trained network learns fundamental spatial- spectral space patterns of neighbouring pixel relations, spectral band correlation, texture and structure that are common across different hyperspectral tasks. We can use this pre-trained feature extractor

as a base for any type of machine learning task and then fine tune it for any downstream tasks [21].

This Thesis investigates the development of Pre-trained deep learning networks for analysing hyperspectral image data. We employ a transformer architecture specifically adapted to hyperspectral data with factorized spatial-spectral attention mechanisms, designed specifically to process high-dimensional hyperspectral data and also compare two prominent self-supervised learning methods are compared: Masked Autoencoding (MAE) [10], which learns representations by reconstructing randomly masked image patches, and Contrastive Learning (SimCLR) [3], which learns discriminative features by distinguishing augmented views of the same image from different samples. Our evaluation identifies which self-supervised objective yields representations that transfer most effectively to downstream spatial-spectral classification tasks under computational and data constraints.

1.2 Research Questions

The research questions defined here guide the investigation of self-supervised pre-training strategies applied to hyperspectral image tasks with limited labelled data. [30] Introduced Masked Vision Transformer architecture specifically designed for hyperspectral image, which showed effectiveness of Masked Autoencoding for learning spatial - spectral features. However this work exclusively focused on reconstruction-based learning through MAE, leaving the alternative approaches unexplored with the same architecture. We investigate whether Contrastive learning as an alternative strategy. We investigate whether this learning objective which optimises by discrimination rather than reconstruction better aligns with our downstream classification tasks

Thus, the following research questions are identified to guide this study :

1. Can self-supervised pre-training improve hyperspectral image classification performance in small-data regimes compared to training from random initialization?
2. Which self-supervised pre-training approach is more effective for hyperspectral classification: reconstruction-based learning (Masked Autoencoding) or discriminative learning (Contrastive Learning)?
3. Do the learned representations from self-supervised pre-training capture meaningful spatial-spectral patterns that transfer effectively to downstream classification tasks?
4. How does the volume of unlabeled pre-training data affect the downstream performance of reconstruction-based and discriminative self-supervised approaches, and at what point do these methods reach a performance plateau?

1.3 Main Contributions

The research led to the following contributions made during this thesis work, which are listed below:

1. We implemented a Vision Transformer architecture with spatial and spectral attention mechanism specifically tailored for processing high-dimensional hyperspectral data with 256 spectral bands.
2. We compare two state of the art self - supervised pre-training methods - Masked Autoencoding (MAE) and Contrastive Learning (SimCLR).
3. We demonstrate that self-supervised pre-training improves downstream hyperspectral classification performance over the corresponding randomly initialized MaskedSST baseline, with MAE providing the strongest transfer performance in the evaluated experiments.

1.4 Thesis Structure

The thesis has the following structure:

- **Chapter 2** reviews existing literature and contemporary methods relevant to hyperspectral representation learning and transformer architectures.
- **Chapter 3** provides an overview of the fundamental concepts of different topics and the background information for the methods used in this work.
- **Chapter 4** details the proposed methodology, multi-domain dataset preprocessing, and the architecture of the Masked Spatial-Spectral Transformer.
- **Chapter 5** describes the software and hardware infrastructure, distributed training setup, optimization schedules, and training configurations
- **Chapter 6** presents the evaluations, covering pre-training convergence, patch size comparisons, performance benchmarks and ablation studies on data volume and domain impact
- **Chapter 7** summarizes the main findings of this research and discusses prospective directions for future work

2

Related Work

This chapter reviews the literature foundational to this thesis, structured into four principal domain areas:

- **Section 2.1** Deep learning architectures for hyperspectral image classification,
- **Section 2.2** Transformer adaptations tailored for spectral-spatial feature extraction,
- **Section 2.3** Self-supervised pre-training paradigms in general computer vision, and
- **Section 2.4** Self-supervised representation learning applied to hyperspectral imaging.

Finally, we synthesize these perspectives to explicitly define the research gap addressed by this work.

2.1 Deep Learning for Hyperspectral Image Classification

Deep learning has become the dominant approach for hyperspectral image (HSI) analysis, largely replacing traditional machine learning techniques such as Support Vector Machines (SVMs) and Random Forests. Paoletti et al. [25] provide a comprehensive review of deep learning classifiers for hyperspectral imaging, tracing the architectural evolution from simple multi-layer perceptrons to sophisticated convolutional and recurrent neural networks.

Early deep learning approaches processed hyperspectral data along a single dimension. Hu et al. [15] introduced 1D Convolutional Neural Networks (CNNs) that treat each pixel’s spectral signature as a 1D sequential signal, applying convolutions along the spectral axis to extract feature representations. While computationally efficient, 1D CNNs completely discard spatial context—neighboring pixels, which frequently share material properties and exhibit correlated reflectance, are processed independently. This spatial blindness severely limits classification accuracy in complex scenarios where local spatial texture carries critical discriminative information.

To incorporate spatial context, Chen et al. [5] applied 2D CNNs to local spatial patches extracted around each target pixel. By processing spatial neighborhoods, these models effectively capture local geometry, edges, and texture patterns. However, the spectral dimension in 2D CNN setups is typically reduced prior to feature extraction using Principal Component Analysis (PCA) or treated as independent channels. This strategy collapses or discards critical spectral correlations necessary for fine-grained material discrimination.

Li et al. [19] addressed both limitations simultaneously by proposing 3D CNNs that process

spatial and spectral dimensions jointly using volumetric convolutional kernels. A 3D kernel of size $k \times k \times d$ operates over local spatial neighborhoods across d consecutive spectral bands, enabling filters to learn coupled spatial-spectral representations. Despite their superior discriminative performance, 3D convolutions introduce significant parameter overhead and computational complexity that scale rapidly with the number of spectral bands, making them exceptionally expensive to train on high-dimensional hyperspectral cubes.

2.2 Transformer Architectures for Hyperspectral Data

The success of Vision Transformers (ViTs) [7] in general computer vision motivated their adoption in hyperspectral data processing. Unlike CNNs, which rely on local receptive fields, transformers leverage global self-attention mechanisms, allowing tokens (patches) across the entire input grid to interact directly regardless of spatial distance. This capability is particularly valuable for capturing long-range spatial context and non-local dependencies in hyperspectral scenes.

Hong et al. [14] extended transformer architectures to HSI with *SpectralFormer*, which applies self-attention mechanisms directly along the spectral dimension to capture cross-band dependencies. SpectralFormer demonstrated that global spectral modeling effectively captures distant band correlations that local 1D convolutional operations overlook—such as relationships between visible absorption features and short-wave infrared (SWIR) reflectance characteristics. However, SpectralFormer primarily operates within the spectral domain and does not fully integrate joint 3D spatial-spectral attention.

Most directly relevant to this thesis, Scheibenreif et al. [30] proposed a Masked Vision Transformer architecture specifically tailored for hyperspectral image classification. Their work introduced factorized spatial-spectral attention modules and demonstrated the effectiveness of Masked Autoencoding (MAE) for learning robust representations from high-dimensional HSI data. Their results established that self-supervised pre-training via masked reconstruction significantly improves downstream classification performance, particularly in data-scarce regimes. However, their study exclusively investigated generative reconstruction (MAE), leaving open the fundamental question of whether contrastive self-supervised strategies could yield equivalent or superior feature representations.

2.3 Self-Supervised Learning in Computer Vision

Self-supervised learning (SSL) has emerged as a key paradigm for learning rich visual representations from unannotated data. Liu et al. [21] categorize SSL into generative approaches (which learn by reconstructing missing or corrupted inputs) and discriminative/contrastive approaches (which learn by maximizing similarity between augmented views of identical instances). Both paradigms significantly reduce the reliance on human-annotated labels for downstream transfer tasks.

In the generative domain, He et al. [10] introduced Masked Autoencoders (MAE), adapting

masked language modeling principles to computer vision. MAE randomly masks a high proportion (75–85%) of image patches and trains an encoder-decoder architecture to reconstruct original pixel values across masked tokens. The high masking ratio prevents shortcut learning via local interpolation, forcing the network to acquire deep structural and semantic representations. Furthermore, because the heavy encoder only processes the remaining unmasked tokens, MAE pre-training achieves high computational efficiency. In the contrastive domain, Chen et al. [3] introduced SimCLR, a framework that learns representations by maximizing agreement between differently augmented views of the same image via the Normalized Temperature-scaled Cross-Entropy (NT-Xent) loss. By contrasting positive pair views against negative samples within a batch, SimCLR encourages the encoder to retain invariant global semantics while discarding task-irrelevant augmentations.

2.4 Self-Supervised Learning in Hyperspectral Imaging

Applying self-supervised learning to hyperspectral imaging presents unique challenges and opportunities due to the physical nature of 3D spatial-spectral data. Recent studies have attempted to adapt general vision SSL paradigms to HSI by treating spectral signatures or spatial patches as separate augmentation targets.

Generative SSL methods in HSI predominantly focus on masked band reconstruction or spatial patch completion, exploiting spatial and spectral redundancy to learn underlying material signatures. Conversely, contrastive SSL methods in HSI rely on domain-specific data augmentations, such as random spectral band dropping, spatial jittering, or Gaussian noise addition, to construct positive and negative pairs. However, defining effective contrastive augmentations for HSI remains non-trivial, as aggressive spatial or spectral transformations risk destroying delicate diagnostic spectral absorption features essential for material classification.

2.5 Research Gap

Existing literature demonstrates the power of deep learning and spatial-spectral transformers for hyperspectral classification, as well as the efficacy of self-supervised pre-training in computer vision. However, direct empirical comparisons of different self-supervised learning paradigms within the same spatial-spectral transformer backbone remain limited, particularly under a controlled multi-domain pre-training setting:

1. **Generative vs. Contrastive Comparison:** Prior benchmark studies in HSI, such as [30], focus almost exclusively on reconstruction-based (MAE) pre-training. Prior work has largely emphasized reconstruction-based pre-training, while fewer studies directly compare reconstruction-based and contrastive objectives within the same hyperspectral transformer backbone and experimental setting.
2. **Multi-Domain Representation Learning:** Many existing HSI self-supervised studies focus on individual scenes, domains, or sensors. In comparison, the use of a uni-

fied spatial-spectral architecture for multi-domain pre-training across heterogeneous hyperspectral datasets remains comparatively less explored.

This thesis addresses these gaps by implementing both MAE and SimCLR objectives within a unified 3D Masked Spatial-Spectral Transformer, conducting a comparative analysis of representation quality and transfer efficiency across diverse evaluation domains.

3

Background

This chapter provides an overview of the core theoretical concepts foundational to this thesis. We review hyperspectral image classification, self-supervised learning paradigms, and Vision Transformer (ViT) architectures. Furthermore, we examine specific self-supervised frameworks—namely, Masked Autoencoding and Contrastive Learning and their direct application to spatial-spectral data. Finally, we explore pre-training strategies for mitigating labeled data scarcity and outline standard evaluation protocols for assessing feature transferability.

The remainder of this chapter is organized as follows:

- **Section 3.1** introduces the fundamentals of hyperspectral imaging (HSI), including physical reflectance properties, spectral characteristics, and data structure.
- **Section 3.2** reviews deep learning architectures applied to hyperspectral data.
- **Section 3.3** introduces Vision Transformers and their adaptation to hyperspectral sequences, detailing patch formation, spatial-spectral multi-head attention, and residual structures.
- **Section 3.4** details self-supervised learning paradigms, examining Masked Autoencoding and Contrastive Learning alongside their loss formulations.
- **Section 3.5** outlines downstream evaluation metrics, including linear probing.

3.1 Fundamentals of Hyperspectral Imaging

Hyperspectral Imagery is an advanced sensing technology that stores detailed information across hundreds of contiguous wavelengths and is one of the most significant discoveries in remote sensing imaging technology. Unlike conventional RGB sensors which capture only 3 broad color channels (red, green, blue) which lie in visible spectrum (400-700 nm), hyperspectral sensors acquire spectral measurements across 100 to 300 + color channels with narrow bandwidths (typically 5-10 nm), spanning wavelengths from the visible (VIS) through near-infrared (NIR, 700-1000 nm) to short-wave infrared (SWIR, 1000-2500 nm) regions of the electromagnetic spectrum[25].

Hyperspectral images creates a rich 3 dimensional data structure called a hyperspectral cube, which can be represented in mathematical form as $I \in \mathbb{R}^{H \times W \times C}$, where H

and W denote the spatial height and width in pixels, and C represents the number of spectral bands. Each pixel at a spatial location (x,y) Contains a spectral signature $s(x,y) = [r_1, r_2, \dots, r_C]$ where r_i represents the reflectance or radiance value at wavelength λ_i .

This spectral signature is unique electromagnetic fingerprint that characterizes the material properties at that spatial location. For example an typical hyperspectral image of 512×512 pixels and 256 spectral bands contains over 67 million individual measurements ($512 \times 512 \times 256 = 67,108,864$) values which provides detailed information of the composition of the target space.

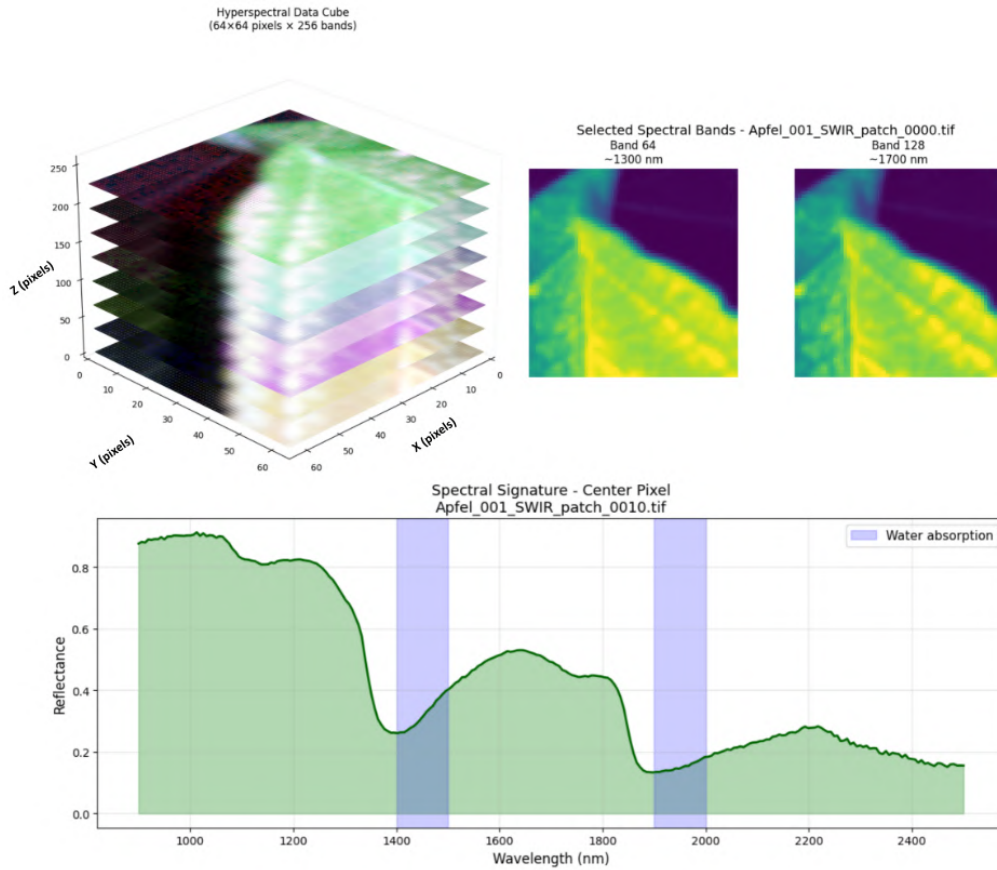


Figure 3.1: A 64x64 patch of hyperspectral data and Spectral Signature from our data

The fundamental principle of capturing reflectance of light at different wavelengths is that each material has different composition, chemical bonds and physical structure and pattern which is unique to this substance[18]. Visible and Near-Infrared (VNIR) sensors operate in the wavelength range of 400-1000 nm, covering the visible spectrum (400-700 nm) and extending into the near-infrared region. Short-Wave Infrared (SWIR) sensors operate in the 1000-2500 nm wavelength range, providing complementary information about molecular composition, moisture content, and chemical properties invisible to VNIR sensors[18]. For instance, Different coffee varieties—Arabica and Robusta—differ in their chemical composition (caffeine content, lipid profiles, organic acids), producing subtle but

measurable differences in their SWIR spectral signatures that are imperceptible to RGB cameras but detectable with hyperspectral sensors.

Hyperspectral imaging presents significant challenges. The sensors are expensive, often costing tens to hundreds of thousands of dollars, limiting accessibility. Data collection requires careful calibration, atmospheric correction for outdoor sensing and controlled illumination. Then there is the size of HSI data as they can be several megabytes to gigabytes easily their storage, transmission and processing becomes a major challenge. Most critically for machine learning applications, acquiring labeled training data is expensive and time-consuming, requiring domain expertise to annotate materials, species, or conditions based on spectral signatures rather than visual appearance alone.

3.2 Deep Learning for Hyperspectral Images

Deep learning techniques have significantly advanced HSI analysis by learning complex feature representations directly from raw spectral data without relying on manual feature engineering.

Early deep learning approaches relied on spectral-only processing. Hu et al. [15] introduced 1D Convolutional Neural Networks (1D-CNNs) that treat each pixel’s spectral signature as a 1D sequence, applying convolutions across the spectral dimension to extract band-wise patterns. While computationally efficient, this formulation entirely neglects spatial context, ignoring valuable structural information provided by neighboring pixels.

To incorporate spatial context, Makantasis et al. [23] applied 2D-CNNs to spatial patches extracted from HSI data. To manage high dimensionality, dimensionality reduction techniques like Principal Component Analysis (PCA) were often applied beforehand to compress the spectral bands. While 2D-CNNs effectively capture localized spatial textures and patterns, relying heavily on PCA can result in the loss of critical, subtle spectral signatures needed for fine-grained discrimination.

To overcome the limitations of separate processing, Chen et al. [5] proposed 3D-CNNs, which jointly model spatial and spectral dimensions using 3D convolutional kernels. A 3D kernel of size $k \times k \times d$ processes a local spatial patch ($k \times k$ pixels) across d consecutive spectral bands simultaneously, learning joint spatial-spectral features. However, 3D convolutions incur substantial computational overhead and high memory demands that scale linearly with the number of spectral channels.

More recently, Transformer architectures have emerged as a promising alternative for HSI processing. Building on the success of Vision Transformers in computer vision, Hong et al. [14] introduced SpectralFormer. Unlike CNNs, which are restricted to local receptive

fields, Transformers leverage global self-attention mechanisms. This allows every spectral band to interact with all other bands across long sequences, capturing non-local spectral dependencies that localized convolutions typically miss.

3.3 Vision Transformer Architecture

The Vision Transformer (ViT) architecture, introduced by Dosovitskiy [7], marked a fundamental shift in computer vision from standard convolutional networks toward transformer-based architectures. Unlike Convolutional Neural Networks (CNNs), which process images via localized convolution operations, Vision Transformers divide an image into a sequence of patches and apply global self-attention. This enables every image region to interact directly with all other regions regardless of spatial distance, allowing the model to capture complex, long-range dependencies effectively.

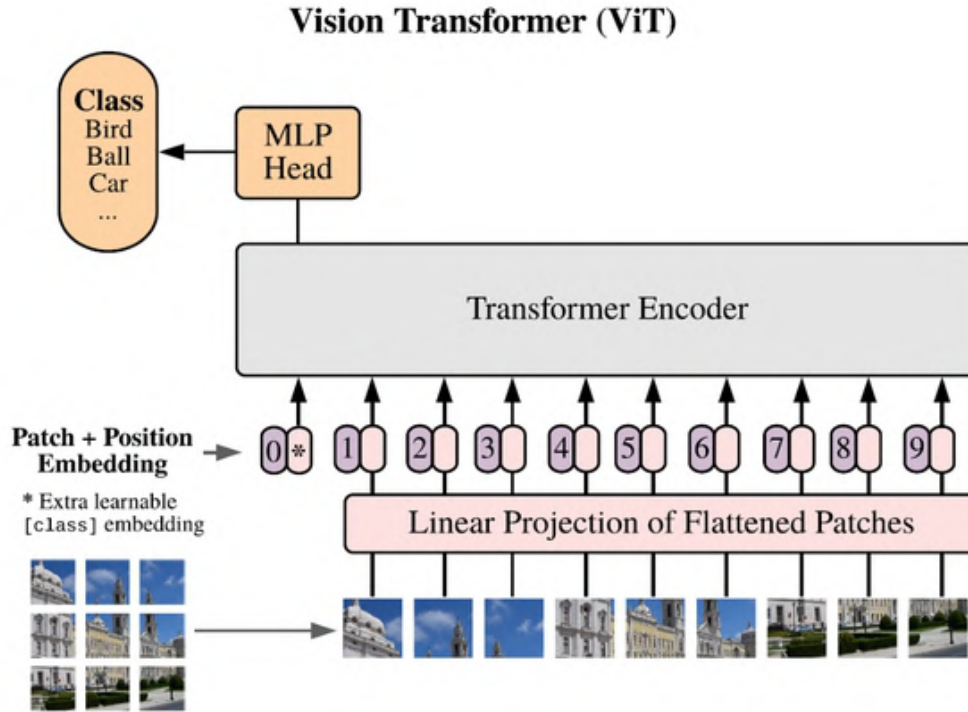


Figure 3.2: Vision Transformer Architecture [7]

3.3.1 Patches Formation

The innovation of the vision transformers lies in the simple architecture and usage of the already used transformer model which has proven successful for language modeling. Like in the transformer model instead of word tokens vision transformer uses image patches, the image of dimensions $H \times W \times C$ (height, width, channels) is treated as N flattened patches. For example, the first stage converts a 2D image into a sequence of patch embeddings

suitable for transformer processing. An input image of size $H \times W \times C$ is divided into non-overlapping patches of fixed size $P \times P$, yielding

$$N = \frac{H \times W}{P^2}$$

patches. To obtain $N = 9$ patches, the image must be partitioned into a 3×3 grid, meaning

$$P = \frac{H}{3}, \quad P = \frac{W}{3}.$$

For instance, an image of size 300×300 with $C = 3$ channels (RGB) divided into 100×100 patches produces $N = 9$ patches, each containing

$$P^2 \times C = 100^2 \times 3 = 30,000$$

values.

3.3.2 Patch Flattening and Linear Projection

Each $P \times P \times C$ patch is flattened into a single vector of length P^2C . These flattened patch vectors are then linearly projected to a fixed latent dimension D through a learnable projection matrix

$$W_{\text{patch}} \in \mathbb{R}^{(P^2C) \times D}.$$

This projection step plays a few important roles. First, it ensures that every patch—no matter its original size—gets represented using the same dimensionality D . It can also reduce the raw patch dimensionality when $D < P^2C$ (e.g., projecting a 30,000-dimensional raw patch to a 3000-dimensional embedding). Beyond that, the learnable projection matrix helps the model highlight useful or discriminative patterns within each patch. After this transformation, we end up with N patch embeddings, each of size D , forming a sequence matrix

$$X \in \mathbb{R}^{N \times D}.$$

ViT also adds a token called CLS token in the start of the of the patch sequence, it gathers information throughout the self-attention process. After passing through layers of the transformers this CLS token becomes the final image-level representation used for classification

3.3.3 Positional Encoding

ViT includes learnable positional embeddings so that the position of the patch where it comes from is known. These embeddings form a matrix $E_{\text{pos}} \in \mathbb{R}^{(N+1) \times D}$, with one entry for every patch and one for the CLS token. These embeddings are added directly to the patch embedding in the input layer and is trained along with the rest of the model. Instead of using a fixed sinusoidal encodings we have learnable encodings so that the transformer can learn the positional patterns, it helps in capturing the 2D layout of the image

patches

3.3.4 Transformer Encoder

The Transformer encoder consists of many identical layers stacked one after another. Each block contains two main components: a Multi-Head Self-Attention (MSA) module and a Feed-Forward Network (FFN). Both these submodules are connected with residual connections and followed by layer normalization ensuring stable optimization and effective gradient flow.

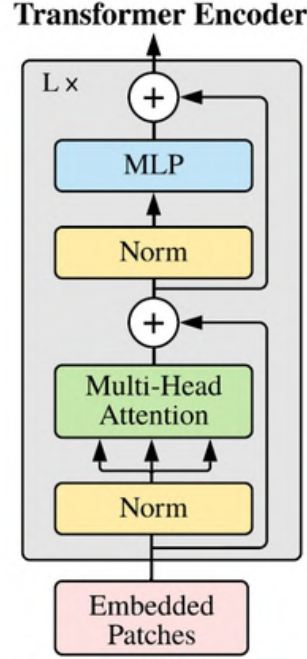


Figure 3.3: Transformer Encoder
[7]

Multi-Head Self-Attention (MSA), Self-attention enables each token to aggregate information from all other tokens in the sequence, with the model learning how strongly each token should influence the others. Given an input sequence

$$Z \in \mathbb{R}^{(N+1) \times D},$$

the model computes queries, keys, and values using learned projection matrices

$$W_Q, W_K, W_V \in \mathbb{R}^{D \times D_{\text{head}}}.$$

These projections are defined as

$$Q = ZW_Q, \quad K = ZW_K, \quad V = ZW_V.$$

The attention mechanism measures the compatibility between each query and all keys

using scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{D_{\text{head}}}}\right)V.[31]$$

The scaling factor $\sqrt{D_{\text{head}}}$ prevents very large dot-product values, which would push the softmax into regions with very small gradients. The softmax operation converts raw similarity scores into normalized attention weights that determine how much each value vector contributes to the output.

Multi-Head, Instead of computing a single attention operation, the transformer uses h parallel attention heads. Each head has its own set of projection matrices, allowing the model to learn different types of relationships simultaneously. For example, one head may focus on spatially adjacent patches, another on semantically similar regions, and others on more abstract or task-specific patterns.

For head i , the computation is

$$\text{head}_i = \text{Attention}(ZW_{Q_i}, ZW_{K_i}, ZW_{V_i}).$$

The outputs of all heads are concatenated and projected back to the model dimension D using a final output matrix:

$$\text{MSA}(Z) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W_O,$$

where

$$W_O \in \mathbb{R}^{(hD_{\text{head}}) \times D}.$$

This projection mixes information across heads and produces the refined token representations that are passed to the subsequent feed-forward network[31].

Position-wise Feed-Forward Network (FFN)

After self-attention aggregates information across all tokens, each token representation passes independently through a two-layer feed-forward network. This network applies the same learned transformation to every token position, hence "position-wise." The FFN is defined as [12]:

$$\text{FFN}(x) = \text{GELU}(xW_1 + b_1)W_2 + b_2,$$

where $W_1 \in \mathbb{R}^{D \times D_{\text{FFN}}}$ expands the embedding dimension to D_{FFN} (typically $4D$), and $W_2 \in \mathbb{R}^{D_{\text{FFN}} \times D}$ projects the representation back to the model dimension. This expansion-compression pattern creates what's called a bottleneck architecture. By forcing information through a wider intermediate representation before compressing it back, the network learns rich, non-linear transformations that go beyond what self-attention alone can capture. The GELU (Gaussian Error Linear Unit) activation function introduces non-linearity between these two layers[12]. Unlike the sharp cutoff of ReLU, GELU provides a smoother activation

curve that tends to work better in transformer architectures.

Residual Connections and Layer Normalization

Both the multi-head attention and feed-forward network components are enhanced with two training stabilization techniques: residual connections and layer normalization. The complete encoder block operates as follows:

$$z'_l = \text{MSA}(\text{LayerNorm}(z_l)) + z_l,$$

$$z_{l+1} = \text{FFN}(\text{LayerNorm}(z'_l)) + z'_l.$$

Residual connections (the “ $+z_l$ ” and “ $+z'_l$ ” terms) create shortcut paths that allow gradients to flow directly backward through the network during training. Without these shortcuts, very deep networks suffer from vanishing gradients—the learning signal becomes progressively weaker as it propagates backward through many layers. Residual connections mitigate this issue, enabling transformers to scale to 12, 24, or even more layers while maintaining stable and reliable training.

Layer normalization standardizes the activations across the feature dimension before each sub-layer. This normalization reduces the sensitivity to initialization and helps maintain stable activation magnitudes throughout training [2]. Vision Transformers use what’s called a “pre-norm” configuration, where normalization happens before each operation rather than after (the “post-norm” approach used in the original transformer). This pre-norm design has proven more stable when training very deep vision models.

Stacking Encoder Blocks

The encoder consists of L such blocks stacked sequentially, with each block refining the token representations produced by the previous one. Early layers typically capture low-level visual patterns such as edges, textures, and local color variations, while deeper layers learn higher-level semantic structure, including object parts and global contextual relationships. In hyperspectral imagery, this hierarchical progression may extend from basic spectral absorption features to material-specific signatures and, ultimately, semantic object categories.

3.4 Self-Supervised Learning for Vision

Supervised deep learning models typically rely on extensive annotated datasets, which are costly and labor-intensive to collect—particularly in hyperspectral imaging where dense pixel-level labeling requires domain expertise. Self-supervised learning (SSL) addresses this constraint by deriving supervision signals directly from unlabeled raw data. By training networks on pretext tasks, SSL forces models to extract fundamental structural, spatial, and spectral representations without human annotations. These pre-trained representations can subsequently be transferred to downstream tasks with significantly reduced labeled

training samples. In this section, we examine two prominent self-supervised paradigms: reconstruction-based learning via Masked Autoencoding (MAE) and discriminative feature alignment via Contrastive Learning (SimCLR).

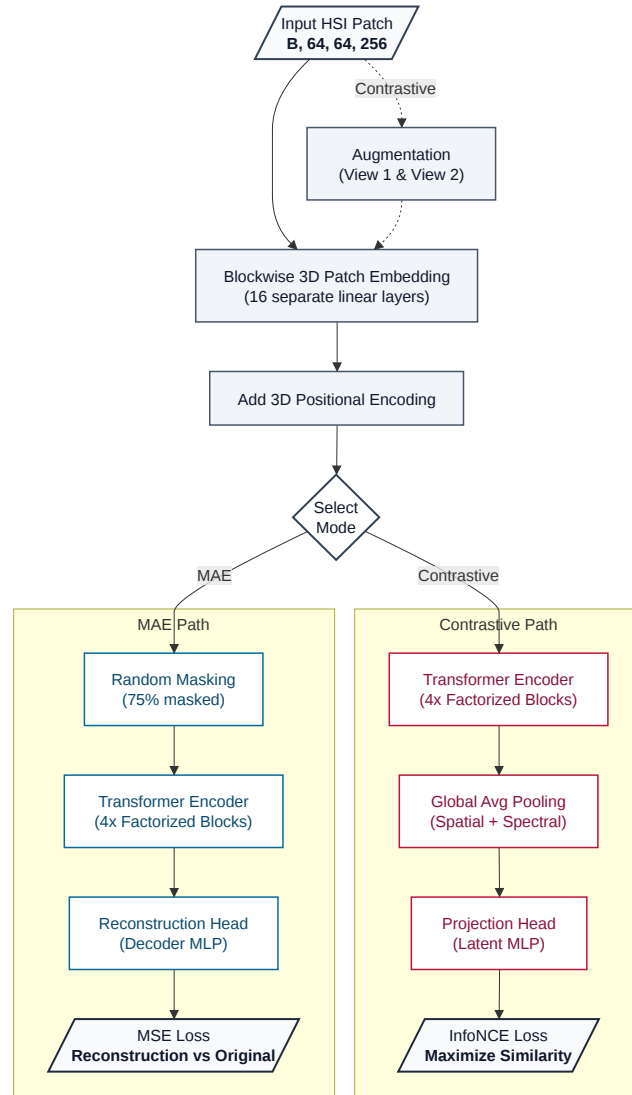


Figure 3.4: Overview of the hyperspectral self-supervised pre-training and downstream fine-tuning pipeline.

3.4.1 Masked Autoencoding

Masked Autoencoding (MAE), introduced by He et al. [10], learns visual representations by reconstructing randomly masked image regions from the visible context. The method adapts masked language modeling from NLP to vision: if a model can accurately reconstruct 75–85% of an image from only 15–25% of visible patches, it must capture spatial-spectral patterns and structural relationships.

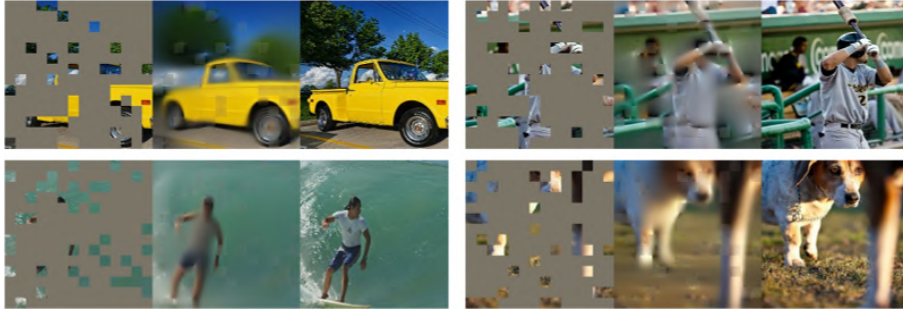


Figure 3.5: MAE reconstruction.[10]

Architecture

MAE employs an asymmetric encoder–decoder design. The encoder, a Vision Transformer (ViT), processes only the visible patches (15–25% of the image), significantly reducing computational cost because self-attention scales quadratically with sequence length. After encoding, learnable mask tokens are inserted at the masked positions, and a lightweight decoder (1–2 transformer blocks) reconstructs the full image. The reconstruction loss is computed using Mean Squared Error (MSE) between predicted and original masked patches.

Training

During pre-training, each image is randomly masked (with a different mask each iteration). The encoder processes visible patches, and the decoder predicts the masked regions. The loss is computed only on masked patches, forcing the model to learn spatial–spectral correlations rather than trivially copying visible content.

3.4.2 SimCLR

SimCLR [3] learns representations by maximizing agreement between augmented views of the same image while minimizing agreement with different images. This discriminative objective aligns naturally with classification tasks: the model learns what makes samples similar or different.

Architecture

SimCLR generates two randomly augmented views of each image, encodes both using a shared Vision Transformer, and applies a projection head (a 2-layer MLP) to map representations into a contrastive embedding space. The NT-Xent (Normalized Temperature-scaled Cross-Entropy) loss pulls positive pairs together while pushing negatives apart:

$$\ell_{i,j} = -\log \left(\frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_k \exp(\text{sim}(z_i, z_k)/\tau)} \right),$$

where $\text{sim}(z_i, z_j)$ denotes cosine similarity and $\tau = 0.5$ is the temperature parameter.

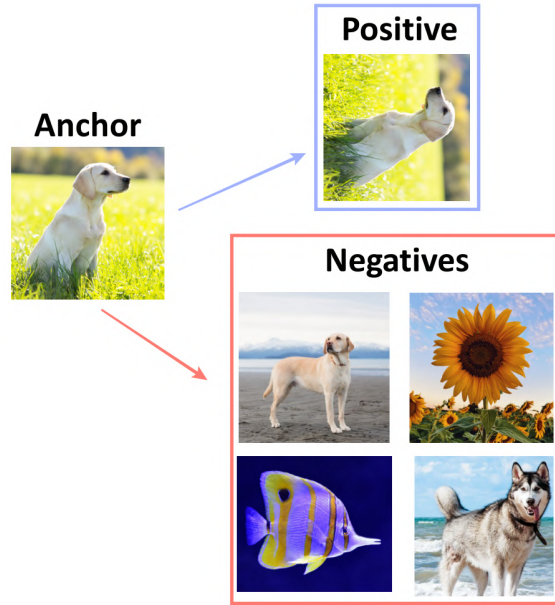


Figure 3.6: Positive and negative sample pairs in contrastive learning .

Data Augmentation

Effective contrastive learning requires strong augmentations that preserve semantic content while altering appearance. For hyperspectral data, suitable augmentations include:

- **Spatial:** Horizontal/vertical flips, rotations, and random cropping.
- **Spectral:** Gaussian noise injection, spectral shifting, band dropout, and spectral scaling.

Training

With a batch size of N , SimCLR creates $2N$ augmented views. Each view forms one positive pair (the alternate augmentation of the same sample) and $2N - 2$ negative pairs (all other images in the batch). Larger batch sizes provide more negatives, improving representation quality but requiring substantial GPU memory.

3.5 Evaluation Protocols for Pre-trained Models

Evaluating the quality of self-supervised pre-trained representations requires protocols that measure how effectively learned features transfer to downstream tasks. Unlike supervised learning where validation accuracy directly reflects model quality, self-supervised methods learn without task-specific objectives, necessitating indirect evaluation approaches. This section reviews standard evaluation protocols used to assess pre-trained feature quality.

3.5.1 Linear Probing

Linear probing evaluates whether pre-trained features are linearly separable for a target task by freezing the encoder and training only a linear classifier on top. Given a pre-trained encoder $f(\cdot)$ that maps images to D -dimensional representations, linear probing adds a single linear layer $W \in \mathbb{R}^{D \times C}$ that maps representations to C classes:

$$y = \text{softmax}(W^\top f(x) + b).$$

During linear probe evaluation, all encoder parameters remain frozen; only the classification layer (W, b) is trained using labeled data. This protocol tests whether the encoder has already learned discriminative features that separate classes without requiring additional feature learning.

Advantages Linear probing provides a fair comparison across different pre-training methods by isolating feature quality from optimization effects. It requires minimal labeled data (hundreds rather than thousands of samples) since only a shallow classifier is trained, reducing the risk of overfitting. Training is computationally efficient (minutes to hours rather than days) due to the small number of trainable parameters. High linear probe accuracy indicates that pre-trained features are task-ready and require minimal adaptation.

Interpretation Strong linear probe performance suggests that the encoder has learned semantically meaningful, discriminative features aligned with the downstream task. Conversely, poor performance indicates that the learned features capture patterns not relevant to the classification objective, implying that substantial fine-tuning may be required for effective adaptation.

3.5.2 Fine-Tuning

Fine-tuning updates all parameters of the pre-trained backbone alongside the task-specific prediction head using downstream labeled data. Unlike linear probing, this approach enables the encoder to adapt its learned representations directly to the target task. Fine-tuning typically employs lower learning rates—often one to two orders of magnitude smaller than those used when training from random weight initialization—to preserve pre-trained feature representations while permitting fine-grained adjustments [8, 10].

Comparison with Linear Probing Analyzing the performance differential between linear probing and fine-tuning provides insight into representation quality and adaptability. A narrow performance gap indicates that the pre-trained features align closely with downstream decision boundaries, requiring minimal adjustment. Conversely, a wide gap signifies that substantial weight adaptation is necessary, which may suggest that the pre-training objective yielded features less directly transferable to the specific downstream task.

3.5.3 Convergence Speed Analysis

Beyond final classification accuracy, convergence speed measures how rapidly models reach peak performance during downstream training. Pre-trained networks with high-quality representations typically converge significantly faster than models initialized from scratch, requiring fewer training epochs and fewer labeled instances to reach target performance metrics. Tracking validation accuracy across training epochs allows us to analyze:

- **Epochs to Convergence:** The number of training epochs required to reach a specific threshold (e.g., 95% of peak validation accuracy).
- **Data Efficiency:** The volume of labeled samples required to achieve a predefined performance benchmark.
- **Gradient Dynamics:** The stability and magnitude of gradient updates during the initial phases of fine-tuning.

3.5.4 Evaluation Metrics

To rigorously evaluate downstream performance across classification, feature representations, and transfer efficiency, the following quantitative metrics are employed:

- **Overall Accuracy (OA):** Measures the proportion of correctly classified hyperspectral samples over the total test samples across all classes:

$$OA = \frac{\sum_{i=1}^C T_i}{N} [27] \quad (3.1)$$

where T_i represents the number of correctly predicted samples in class i , C is the total number of target classes, and N is the total number of evaluation samples.

- **Average Accuracy (AA):** Computes the unweighted arithmetic mean of individual class accuracies to ensure balanced assessment across imbalanced target datasets:

$$AA = \frac{1}{C} \sum_{i=1}^C \frac{T_i}{N_i} \quad (3.2)$$

where N_i denotes the total number of ground-truth samples belonging to class i .

- **Kappa Coefficient (κ):** Evaluates classification agreement corrected for chance agreement, quantifying consistency between predictions and ground-truth labels:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (3.3)$$

where P_o is the observed proportional agreement (OA) and P_e is the expected proportion of agreement under random chance[6].

- **Mean Squared Error (MSE) / L1 Loss:** Used during self-supervised Masked Autoencoding (MAE) pre-training to quantify spectral-spatial reconstruction quality

between original unmasked patch values $x_{s,\lambda}$ and predicted representations $\hat{x}_{s,\lambda}$ across masked token locations $\mathcal{M}$:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{M}|} \sum_{(s,\lambda) \in \mathcal{M}} \|\text{Decoder}(z_{s,\lambda}) - x_{s,\lambda}\|_1 \quad (3.4)$$

- **Normalized Temperature-scaled Cross-Entropy (NT-Xent) Loss:** Used during SimCLR contrastive pre-training to measure feature representation quality by assessing similarity between positive augmented views against negative samples:

$$\ell_{i,j} = -\log \left(\frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_k \exp(\text{sim}(z_i, z_k)/\tau)} \right) \quad (3.5)$$

where $\text{sim}(z_i, z_j)$ denotes cosine similarity in the normalized latent space and τ represents the temperature parameter [4].

4

Design and Development

In Chapter 3, the objective was to provide an overview of the important background related to the concepts for our thesis work. This chapter presents the specific design details and architectural methodology of our approach. The chapter contains the following sections:

- Section 4.1 provides the methodology used to address the research questions stated in Section 1.3.
- Section 4.2 describes the multi-domain hyperspectral dataset, radiometric calibration pipeline, patch extraction strategy, and data standardization.
- Section 4.3 details the MaskedSST architecture, including blockwise 3D patch embedding, learnable positional encoding, random token masking, factorized spatial-spectral attention, and task-specific output heads.
- Section 4.4 provides a summary of the design and development work presented in this chapter.

4.1 Methodology

This section outlines our approach to addressing the research questions posed in Section 1.3. Our methodology follows a systematic workflow: we first aggregate a large-scale, multi-domain hyperspectral dataset to enable robust self-supervised learning, applying necessary calibration and preprocessing to standardize diverse data sources. We then implement a Vision Transformer architecture adapted for hyperspectral data and apply two distinct pre-training strategies—Masked Autoencoding (MAE) and Contrastive Learning (SimCLR)—on this extensive unlabeled dataset. Finally, we evaluate the quality of the learned representations by fine-tuning the pre-trained encoder on a specific downstream task: variety classification.

The experimental design leverages the "pre-train once, deploy anywhere" paradigm. By pre-training on a massive collection of 4,512 hyperspectral samples across 6 domains—ranging from biological samples like apple leaves and coffee beans to materials like paper and pottery—we aim to learn a universal feature extractor that captures fundamental spatial-spectral principles. This approach directly addresses the data scarcity challenge in agricultural applications, where labeled task-specific data is limited, but diverse unlabeled hyperspectral data is abundant.

4.2 Dataset and Data Preprocessing

4.2.1 Multi-Domain Pre-training Dataset

To develop a robust and generalizable pre-trained model, we constructed a multi-domain hyperspectral dataset aggregation comprising 4,512 spatial-spectral patches from six distinct application domains. To ensure architectural consistency across heterogeneous domain sources, all inputs are standardized to spatial patches of 64×64 pixels, with their spectral channels uniformly aggregated to 256 bands. Our aggregated corpus includes:

Table 4.1: Overview of Hyperspectral Datasets

Dataset (Identifier)	Description	No. of Sub-domains	No. of Images
Coffee Beans	Robusta, Arabica	2	2000
Apple Leaves	Apple leaves from different trees	4	1500
Paper Materials	Material identification scans	1	200
Sugar Compounds	Pure chemical endmembers (D-Galactose, D-Glucose, D-Fructose)	3	560
Archaeological Artifacts	Pottery shards with mineralogical signatures	1	150
Grape Samples	Grape fruit and plant vegetation data	1	102
Total	Multi-Domain Corpus	12	4512

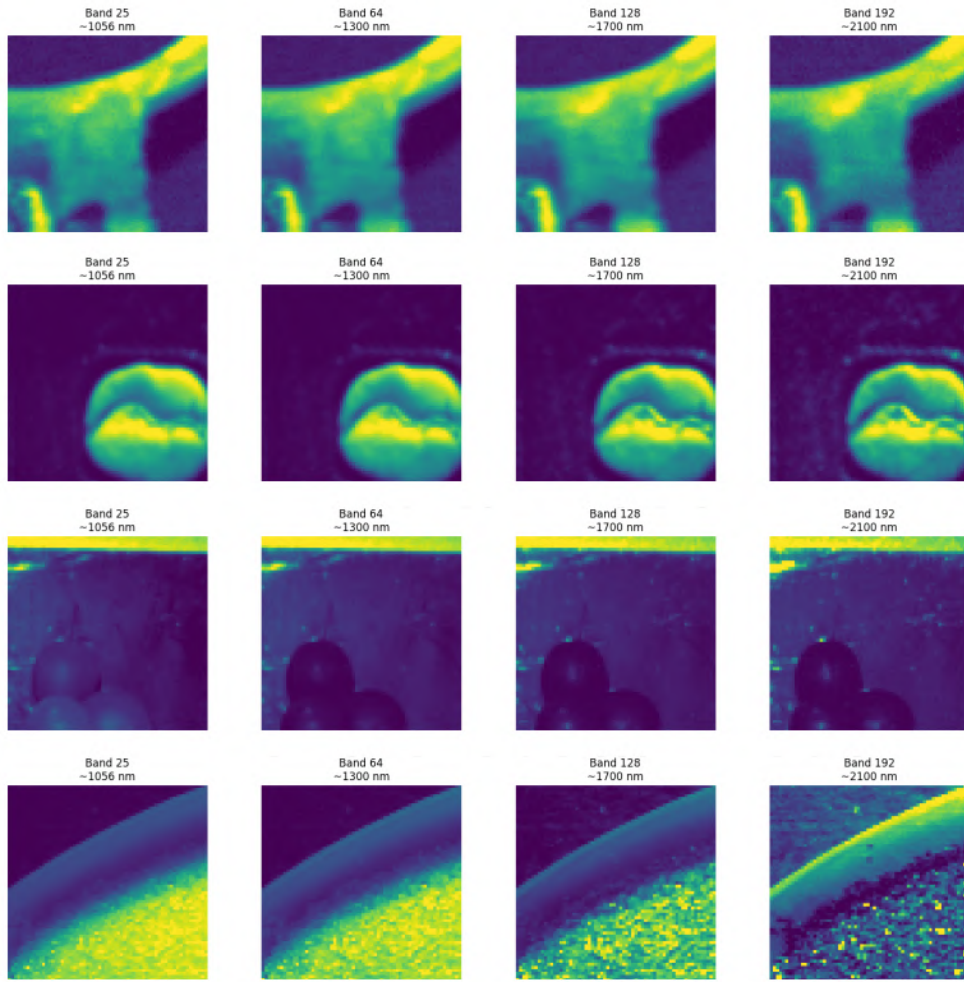


Figure 4.1: 64×64 pixel image patches of Archaeological Artifacts, Coffee Beans, Sugar Compounds, and Grape Samples.

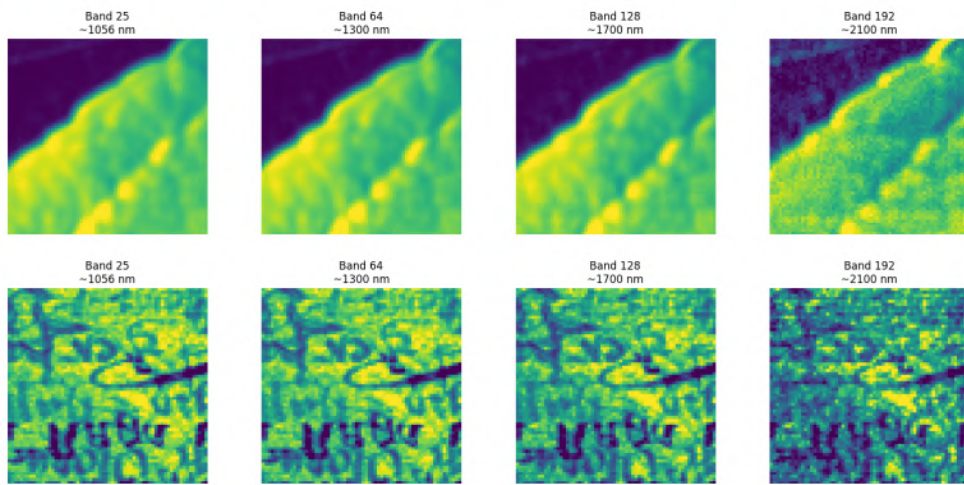


Figure 4.2: 64×64 pixel image patches of Apple Leaves and Paper Materials.

This diversity is critical: by exposing the model to both biological (leaves, beans, grapes) and non-biological (paper, pottery, sugar) materials, the self-supervised objectives force the network to learn fundamental principles of hyperspectral imaging—such as spectral continuity, edge correlations across bands, and material-invariant spatial patterns—rather than overfitting to specific crop characteristics.

4.2.2 Data Calibration and Standardization

A major challenge in aggregating diverse hyperspectral sources is the inconsistency in raw data formats. While the coffee and sugar datasets were already processed, the raw Apple (Apfel) and archaeological datasets required rigorous radiometric calibration to ensure physical consistency.

Calibration Pipeline

Raw hyperspectral sensors record digital numbers (DN) representing light intensity, which depend on illumination and sensor sensitivity. To convert these to physical reflectance (0-1 range), we implemented a calibration pipeline using white reference standards. For each raw image:

1. **White Reference Extraction:** We identified specific spatial regions in the raw scans containing white reference panels (e.g., SWIR rows 0-320, columns 385-651). The median spectrum of this region $I_{white}(\lambda)$ serves as the illumination profile.
2. **Radiometric Calibration:** The reflectance $R(x, y, \lambda)$ at each pixel was computed by normalizing the raw radiance $I_{raw}(x, y, \lambda)$ against the white reference[9]:

$$R(x, y, \lambda) = \frac{I_{raw}(x, y, \lambda)}{I_{white}(\lambda)} \quad (4.1)$$

3. **Spectral Smoothing:** To reduce sensor noise, we applied a Savitzky-Golay filter[29] (window length 11, polyorder 3) to the white reference spectrum before division.
4. **Percentile Normalization:** Finally, reflectance values were normalized by their 99th percentile to correct for specular reflections and mapped to the $[0, 1]$ interval.

This calibration ensures that spectral signatures across all domains represent true material reflectance, allowing the model to learn valid physical correlations.

4.2.3 Patch Extraction and Size Selection

All calibrated images were processed into uniform patches for model training. We selected a patch size of 64×64 pixels based on a strategic trade-off between spatial context and computational efficiency.

Justification for 64×64 Patch Size:

1. **Spatial Context:** A 64×64 window covers sufficient surface area to capture meaningful texture patterns (e.g., leaf venation, bean curvature) and object boundaries. Smaller patches (e.g., 32×32) often fall entirely within homogeneous regions, reducing the self-supervised learning signal (making neighbor prediction trivial).
2. **Spectral Dimensionality Balance:** With 256 spectral bands, a 64×64 patch creates a data cube of $64 \times 64 \times 256 \approx 10^6$ values. This volume provides a rich signal for training transformer models without overwhelming GPU memory, which typically occurs with larger patches (e.g., 128×128).
3. **ViT Compatibility:** Vision Transformers process images as sequences of smaller sub-patches (tokens). A 64×64 image divides cleanly into 16×16 tokens of size 4×4 , creating a sequence length of 256 tokens—ideal for standard transformer attention mechanisms which have quadratic complexity [31] $O(N^2)$.

4.2.4 Data Format and Preprocessing

Patches were saved as GeoTIFF files to maintain metadata and enable efficient parallel loading. The final dataset consists of 4,512 total GeoTIFF files.

Preprocessing Pipeline:

- **Spectral Band Aggregation:** To handle spectral dimension heterogeneity across different datasets, original spectral bands were aggregated and resampled into a uniform target of 256 channels.
- **Pixel Normalization:** We apply pixel-wise normalization across the spectral dimension to standardize continuous intensity signatures. Given an individual pixel vector $x \in \mathbb{R}^C$ with C spectral channels, we compute $\hat{x} = \frac{x - \mu_x}{\sigma_x + \epsilon}$, where μ_x and σ_x denote the mean and standard deviation across the spectral channels of that specific pixel.
- **Masking:** Each sample includes a validity mask (band 257) indicating foreground regions. The model learns to ignore background pixels during loss computation.
- **Train-Validation Split:** We performed a stratified split at the domain level, randomly assigning 80% of files from *each* of the sub-domains to training and 20% to validation. This ensures the validation set (912 samples) accurately reflects the diversity of the full dataset (3,600 training samples), preventing domain shift issues during evaluation.

4.3 Model Architecture

The backbone architecture used in this work is the Masked Spatial-Spectral Transformer (MaskedSST), originally proposed by Scheibenreif et al. [30] for self-supervised hyperspectral image classification. The framework adapts the Vision Transformer (ViT) paradigm to 3D volumetric hyperspectral cubes ($64 \times 64 \times 256$) through a multi-stage processing pipeline:

- **Preprocessing & Embedding:** Raw hyperspectral cubes undergo radiometric calibration and normalization before being partitioned into non-overlapping $4 \times 4 \times 16$ spatial-spectral 3D patches.
- **Positional Encoding:** A learnable 3D positional encoding matrix is added to the patch embeddings to preserve spatial and spectral layout information.
- **Factorized Encoder:** A factorized spatial-spectral transformer encoder processes the token sequences efficiently by decoupling spatial and spectral attention modules.
- **Downstream Heads:** Dedicated task heads enable both self-supervised reconstruction and downstream classification fine-tuning.

The complete model parameters total **1,652,609** trainable weights.

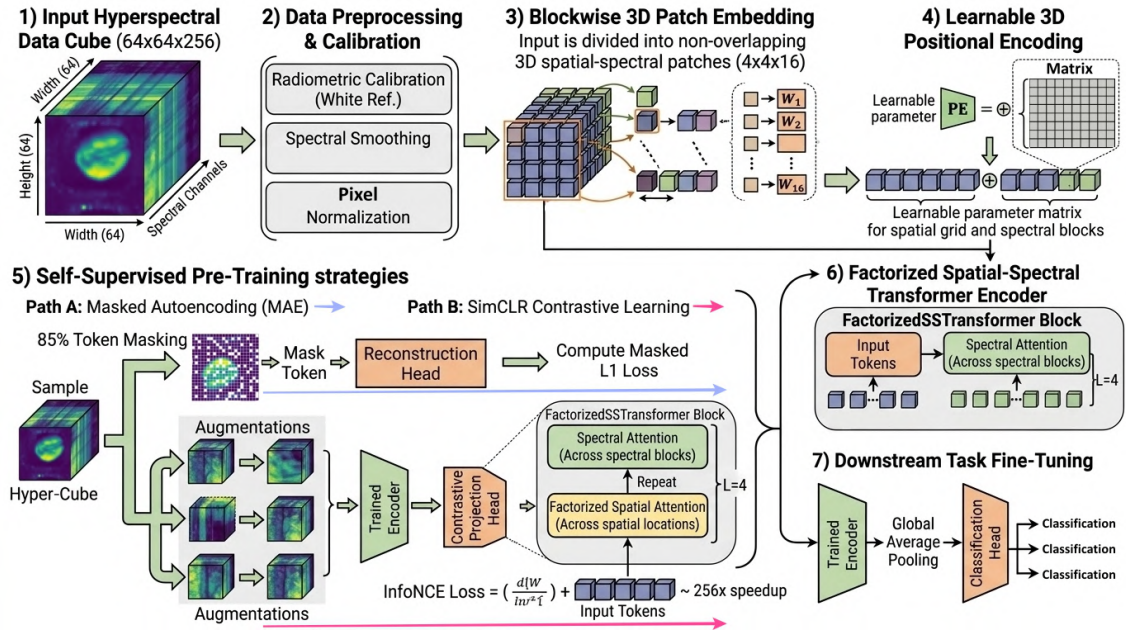


Figure 4.3: Complete architecture pipeline of MaskedSST incorporating pre-processing, 3D patch embedding, self-supervised pre-training (MAE and SimCLR), and downstream fine-tuning.

4.3.1 Blockwise 3D Patch Embedding

Standard patch embedding layers in Vision Transformers use a single convolutional or linear projection matrix applied uniformly across the entire image. However, in hyperspectral imaging, the physical and chemical properties of light vary dramatically across the electromagnetic spectrum. Visible light (400-700 nm) predominantly captures pigmentation and color, near-infrared (700-1400 nm) is sensitive to cellular structure and water content, while short-wave infrared (1400-2500 nm) reveals molecular absorption features related to chemical bonds (C-H, O-H, N-H). Treating all wavelengths identically would obscure these domain-specific characteristics.

Blockwise Patch Embedding

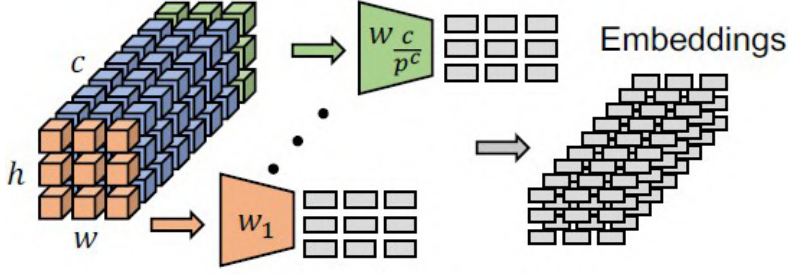


Figure 4.4: Blockwise 3D Patch Embedding Strategy [30]. Input spectral cube is divided into spatial-spectral patches, and each spectral block is projected by a dedicated learnable weight matrix W_c .

To address this, we implemented a **Blockwise Spectral Embedding** strategy inspired by the physical stratification of the spectrum. Given an input hyperspectral cube $X \in \mathbb{R}^{64 \times 64 \times 256}$, we decompose it into non-overlapping 3D patches with dimensions $P_h \times P_w \times P_c = 4 \times 4 \times 16$, where P_h and P_w denote spatial patch dimensions and P_c denotes spectral patch size (number of contiguous bands grouped together). This discretization yields:

- **Spatial Grid:** $N_{spatial} = \left\lfloor \frac{64}{4} \right\rfloor \times \left\lfloor \frac{64}{4} \right\rfloor = 16 \times 16 = 256$ spatial patch locations.
- **Spectral Grid:** $N_{spectral} = \left\lfloor \frac{256}{16} \right\rfloor = 16$ spectral blocks, each representing a contiguous 16-band segment of the spectrum.

The key innovation lies in the projection mechanism. Rather than using a single shared embedding matrix, we instantiate a learnable list of *independent* projection matrices $\{W_1, W_2, \dots, W_{16}\}$, where each $W_\lambda \in \mathbb{R}^{(P_h \cdot P_w \cdot P_c) \times D}$ is dedicated to a specific spectral block λ . For a 3D patch vector $x_{s,\lambda} \in \mathbb{R}^{4 \cdot 4 \cdot 16}$ at spatial location s and spectral block λ , the embedding operation is:

$$z_{s,\lambda}^{(0)} = x_{s,\lambda} W_\lambda + b_\lambda \quad (4.2)$$

where W_λ is the weight matrix for the λ -th spectral block, b_λ is the corresponding bias vector, and $z_{s,\lambda}^{(0)} \in \mathbb{R}^D$ with $D = 128$. This design enables the model to learn distinct feature extractors optimized for different wavelength regimes: early projections (W_1, W_2) learn to extract visible-spectrum features (edges, textures), middle projections (W_7, W_8) specialize in NIR vegetation indices, and later projections (W_{14}, W_{15}, W_{16}) focus on SWIR molecular signatures.

The forward pass through the patch embedding module proceeds as follows:

1. **Reshape:** Transform input $X \in \mathbb{R}^{B \times 64 \times 64 \times 256}$ into $\mathbb{R}^{B \times 16 \times 4 \times 16 \times 4 \times 16 \times 16}$ by grouping spatial and spectral dimensions into patches.
2. **Permute:** Rearrange to $\mathbb{R}^{B \times 256 \times 16 \times 256}$, where the second dimension indexes spatial patches, the third indexes spectral blocks, and the fourth flattens the patch content.

3. **Project:** For each spectral block λ , apply W_λ to all spatial locations simultaneously:
 $\mathbf{Z}_\lambda = \mathbf{X}_{[:, :, \lambda]} W_\lambda$.
4. **Stack:** Concatenate the projected blocks along the spectral dimension to yield
 $\mathbf{Z}^{(0)} \in \mathbb{R}^{B \times 256 \times 16 \times 128}$.

4.3.2 Learnable 3D Positional Encoding

Transformer architectures are inherently permutation-invariant; without explicit positional information, the model cannot distinguish whether two tokens originate from adjacent or distant spatial locations, nor can it resolve the spectral ordering of wavelengths. Unlike standard Vision Transformers that employ fixed sinusoidal positional encodings, we adopt **learnable positional embeddings** that are optimized end-to-end during training.

For the token grid of size $N_{spatial} \times N_{spectral} = 256 \times 16$, we initialize a learnable parameter matrix $\mathbf{PE} \in \mathbb{R}^{16 \times 16 \times 16 \times 128}$, where the first two dimensions correspond to the spatial grid $(i, j) \in \{0, \dots, 15\} \times \{0, \dots, 15\}$, the third dimension indexes the spectral block $\lambda \in \{0, \dots, 15\}$, and the final dimension matches the embedding size $D = 128$. This parameter matrix is initialized randomly from a truncated normal distribution with standard deviation $\sigma = 0.02$. During the forward pass, for each token at spatial location (i, j) and spectral block λ , we retrieve the corresponding positional embedding $\mathbf{PE}_{(i, j, \lambda)} \in \mathbb{R}^{128}$ and add it element-wise to the patch embedding:

$$\tilde{z}_{(i, j, \lambda)}^{(0)} = z_{(i, j, \lambda)}^{(0)} + \mathbf{PE}_{(i, j, \lambda)} \quad (4.3)$$

The key advantage of learnable positional encodings is their flexibility to capture complex, non-local spatial-spectral relationships without assuming rigid distance metrics [14]. Unlike fixed encodings, learnable parameters can adaptively model dependencies between non-adjacent spectral bands and spatial context during pre-training [11].

4.3.3 Random Token Masking

For the Masked Autoencoding (MAE) pre-training objective, we require a masking strategy that removes sufficient information to prevent trivial reconstruction while encouraging the model to learn robust spatial-spectral representations. We implement a **Random Token Masking** strategy that operates independently on each token in the 3D grid.

The masking algorithm proceeds as follows:

1. **Flatten Token Grid:** For each sample in the batch, flatten the 3D token tensor from $\mathbb{R}^{256 \times 16 \times 128}$ to $\mathbb{R}^{N_{total} \times 128}$, where $N_{total} = 256 \times 16 = 4096$ is the total number of tokens.
2. **Random Permutation:** Generate a random permutation π of the token indices $\{0, 1, \dots, 4095\}$. This ensures that each sample in the batch receives a different random mask, enhancing training diversity.

3. **Select Masked Tokens:** Using the masking ratio $\rho = 0.85$, select the first $N_{masked} = \lfloor 4096 \times 0.85 \rfloor = 3481$ indices from the permuted sequence. Create a binary mask $M \in \{0, 1\}^{4096}$ where $M_{\pi(i)} = 1$ for $i \in \{0, \dots, 3480\}$ (masked tokens) and $M_j = 0$ otherwise (visible tokens).
4. **Replace with Mask Token:** For each masked position, replace the learned token embedding with a shared learnable mask token $\mathbf{m} \in \mathbb{R}^{128}$:

$$\tilde{z}_i = \begin{cases} \mathbf{m} & \text{if } M_i = 1 \\ z_i & \text{if } M_i = 0 \end{cases} \quad (4.4)$$

5. **Reshape:** Reshape the masked token sequence back to $\mathbb{R}^{256 \times 16 \times 128}$ for processing by the transformer encoder.
6. **Loss Computation:** The reconstruction loss is computed *only* on the masked tokens. Visible tokens do not contribute to the gradient updates.

This random masking strategy offers several advantages for hyperspectral data. Unlike spatial tube masking (which masks entire vertical columns through all spectral bands), random masking forces the model to learn both spatial *and* spectral reconstruction simultaneously. When a specific spectral band at a specific spatial location is masked, the model must infer the missing value by reasoning about:

- **Spatial context:** Neighboring spatial pixels that share similar material properties.
- **Spectral context:** Adjacent spectral bands at the same location that exhibit correlated reflectance due to physical continuity.
- **Global semantics:** The overall object class (e.g., if surrounding context suggests a coffee bean, the masked region should exhibit typical coffee reflectance signatures).

The high masking ratio of 85% ensures that the reconstruction task remains non-trivial. With only 15% of tokens visible (approximately 614 out of 4096), the model cannot rely on simple interpolation and must develop a deep understanding of hyperspectral material signatures to accurately reconstruct the missing 3481 tokens.

4.3.4 Factorized Spatial-Spectral Transformer Encoder

The total number of tokens for a single hyperspectral patch is $N_{total} = N_{spatial} \times N_{spectral} = 256 \times 16 = 4096$. Computing standard self-attention requires forming an attention matrix of size $N_{total} \times N_{total} \approx 16.8$ million elements, with computational complexity $O(N_{total}^2 \cdot D) \approx 2.15$ billion FLOPs per attention layer. For a 4-layer network with batch size 32, this becomes prohibitively expensive even on modern GPUs.

To resolve this bottleneck, we employ **Factorized Attention**, which decomposes the full 3D attention into two sequential 2D operations: spatial attention followed by spectral attention. Each `FactorizedSSTransformerBlock` applies the following sequence:

Spatial Attention

First, we reshape the token tensor $\mathbf{Z} \in \mathbb{R}^{B \times 256 \times 16 \times 128}$ to treat spectral blocks as part of the batch dimension: $\mathbf{Z}_{spatial} \in \mathbb{R}^{(B \cdot 16) \times 256 \times 128}$. Multi-head self-attention (MHSA) with 8 heads is then applied across the 256 spatial locations:

$$\mathbf{Z}'_{spatial} = \text{LN}(\mathbf{Z}_{spatial}) \quad (4.5)$$

$$\text{Attn}_{spatial} = \text{MHSA}_{spatial}(\mathbf{Z}'_{spatial}, \mathbf{Z}'_{spatial}, \mathbf{Z}'_{spatial}) \quad (4.6)$$

$$\mathbf{Z}_{spatial} \leftarrow \mathbf{Z}_{spatial} + \text{Attn}_{spatial} \quad (4.7)$$

$$\mathbf{Z}_{spatial} \leftarrow \mathbf{Z}_{spatial} + \text{MLP}_{spatial}(\text{LN}(\mathbf{Z}_{spatial})) \quad (4.8)$$

where LN denotes Layer Normalization and $\text{MLP}_{spatial}$ is a two-layer feed-forward network with GELU activation and expansion ratio 4 (hidden dimension 512). This operation allows each spectral block to aggregate information from all spatial locations independently, learning spatial texture patterns.

Spectral Attention

Next, the tensor is reshaped to $\mathbf{Z}_{spectral} \in \mathbb{R}^{(B \cdot 256) \times 16 \times 128}$, treating spatial locations as the batch dimension. MHSA is applied across the 16 spectral blocks:

$$\mathbf{Z}'_{spectral} = \text{LN}(\mathbf{Z}_{spectral}) \quad (4.9)$$

$$\text{Attn}_{spectral} = \text{MHSA}_{spectral}(\mathbf{Z}'_{spectral}, \mathbf{Z}'_{spectral}, \mathbf{Z}'_{spectral}) \quad (4.10)$$

$$\mathbf{Z}_{spectral} \leftarrow \mathbf{Z}_{spectral} + \text{Attn}_{spectral} \quad (4.11)$$

$$\mathbf{Z}_{spectral} \leftarrow \mathbf{Z}_{spectral} + \text{MLP}_{spectral}(\text{LN}(\mathbf{Z}_{spectral})) \quad (4.12)$$

This enables each spatial location to model correlations across wavelengths, capturing spectral signatures (e.g., chlorophyll absorption peaks, water bands).

Complexity Analysis: The factorized approach reduces attention complexity from $O(N_{spatial} \cdot N_{spectral})^2 = O(4096^2) \approx 16.8M$ operations to $O(N_{spatial}^2 + N_{spectral}^2) = O(256^2 + 16^2) \approx 65.8K$ operations per layer—a **256× speedup**. This enables processing of full hyperspectral patches without requiring aggressive downsampling or patch discarding strategies.

The encoder consists of $L = 4$ stacked **FactorizedSSTransformerBlock** layers, followed by a final Layer Normalization. The output is a contextually-enriched token representation $\mathbf{Z}^{(L)} \in \mathbb{R}^{B \times 256 \times 16 \times 128}$ that has integrated both spatial texture information and spectral signature information.

4.3.5 Task-Specific Output Heads

Depending on the training phase and objective, different output heads are attached to the encoder. Our implementation supports three operational modes:

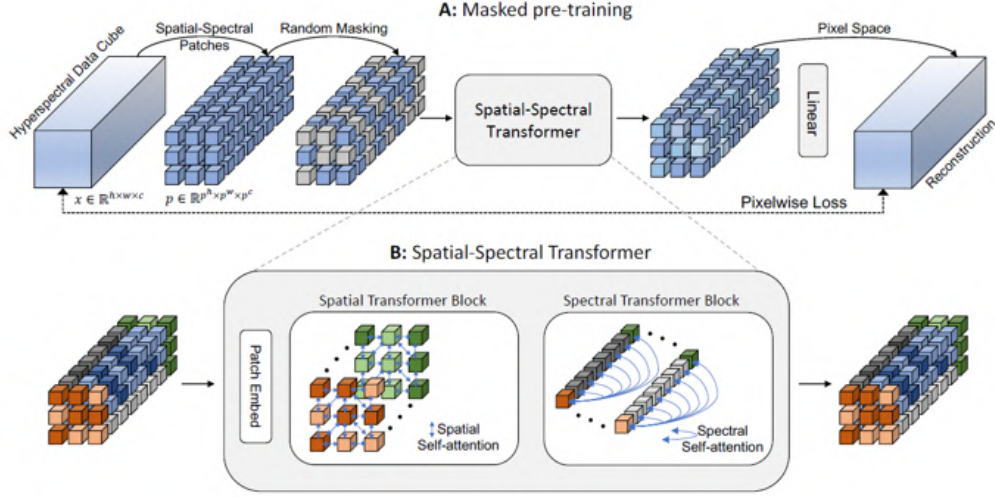


Figure 4.5: Masked Autoencoding (MAE) pre-training framework [30]. (A) The input hyperspectral cube is partitioned into 3D spatial-spectral patches, randomly masked, and processed by the factorized Spatial-Spectral Transformer encoder. A linear head reconstructs the original pixel values evaluated via pixelwise loss. (B) Internal structure of the Spatial-Spectral Transformer block detailing factorized spatial and spectral self-attention mechanisms.

Reconstruction Head (MAE Pre-training)

For Masked Autoencoding, a lightweight decoder reconstructs the original patch values from the encoded representations. The `ReconstructionHead` is implemented as a three-layer MLP:

$$\text{Decoder}(z) = \text{Linear}_{256 \rightarrow 256}(\text{GELU}(\text{LN}(\text{Linear}_{128 \rightarrow 256}(z)))) \quad (4.13)$$

where the output dimension 256 corresponds to the flattened patch size ($4 \times 4 \times 16 = 256$). The decoder operates token-wise on all $256 \times 16 = 4096$ tokens. The reconstruction loss is computed as the masked L1 loss between the reconstructed values and the original patch values:

$$\mathcal{L}_{recon} = \frac{1}{|\mathcal{M}|} \sum_{(s,\lambda) \in \mathcal{M}} \|\text{Decoder}(z_{s,\lambda}) - x_{s,\lambda}\|_1 \quad (4.14)$$

where $\mathcal{M}$ denotes the set of masked tokens.

Classification Head (Fine-tuning)

For downstream classification tasks (e.g., coffee variety identification), the `ClassificationHead` first applies global average pooling over both spatial and spectral dimensions to obtain a single feature vector:

$$v = \frac{1}{N_{spatial} \cdot N_{spectral}} \sum_{s=1}^{256} \sum_{\lambda=1}^{16} z_{s,\lambda} \quad (4.15)$$

This aggregated representation $v \in \mathbb{R}^{128}$ is then passed through a linear classifier:

$$\hat{y} = \text{Softmax}(W_{cls}v + b_{cls}) \quad (4.16)$$

where $W_{cls} \in \mathbb{R}^{128 \times K}$ projects to K class logits (e.g., $K = 3$ for Arabica, Canephora, and Liberica coffee varieties).

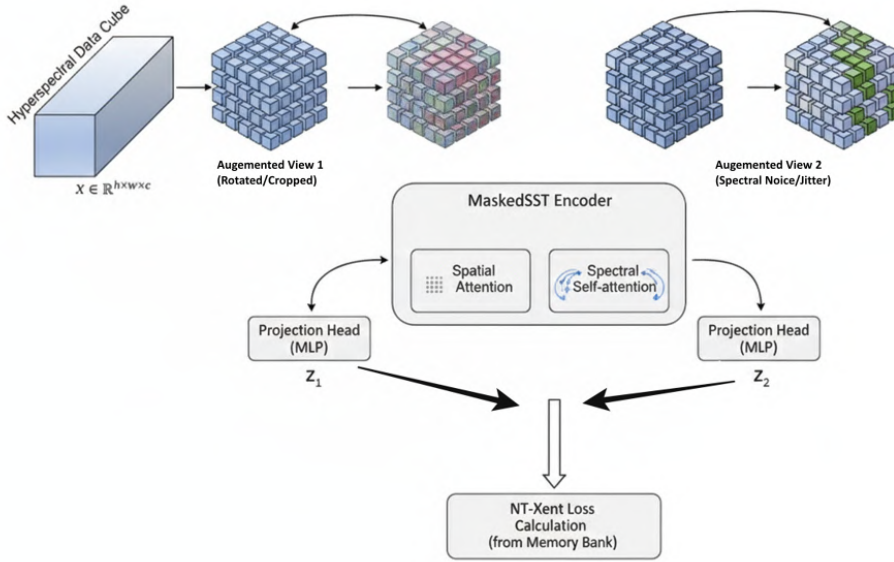


Figure 4.6: SimCLR contrastive pre-training workflow. Two augmented views of the hyperspectral data cube—spatial transformation (e.g., rotation/crop) and spectral jittering—are passed through the shared MaskedSST encoder. The projected representations z_1 and z_2 are optimized using the NT-Xent contrastive loss.

Contrastive Projection Head (SimCLR Pre-training)

For contrastive learning, the `ProjectionHead` maps the encoder output to a compact metric space suitable for contrastive loss computation. After global pooling, a two-layer MLP projects the features to a 128-dimensional unit sphere:

$$z_{contrast} = \text{L2-Normalize}(\text{Linear}_{128 \rightarrow 128}(\text{ReLU}(\text{Linear}_{128 \rightarrow 128}(v)))) \quad (4.17)$$

The InfoNCE loss is then computed between augmented views of the same sample in this

normalized space.

4.3.6 Model Summary and Hyperparameters

The complete MaskedSST architecture is summarized in Table 4.2. The model operates in three distinct modes controlled by the `mode` parameter in the forward pass: `mode="reconstruction"` activates the reconstruction head with random masking for MAE pre-training; `mode="contrastive"` activates the projection head without masking for SimCLR pre-training; and `mode="classification"` activates the classification head without masking for fine-tuning. Weight initialization follows standard Vision Transformer practices: linear layers are initialized with a truncated normal distribution ($\sigma = 0.02$), layer normalization parameters are set to $\gamma = 1$ and $\beta = 0$, and the mask token is initialized to zeros and learned during training.

Table 4.2: MaskedSST Hyperparameters

Component	Configuration
Input Size	$64 \times 64 \times 256$
Spatial Patch Size	$P_h = 4, P_w = 4$
Spectral Patch Size	$P_c = 16$
Spatial Tokens	$N_{spatial} = 256$
Spectral Tokens	$N_{spectral} = 16$
Total Tokens	$N_{total} = 4096$
Embedding Dimension	$D = 128$
Encoder Depth	$L = 4$ layers
Attention Heads	$h = 8$ heads per layer
MLP Expansion Ratio	4
Dropout Rate	0.1
Masking Ratio (MAE)	0.85 (random token masking)
Total Parameters	1,652,609
Trainable Parameters	1,652,609

4.4 Summary

This chapter presented the methodology, dataset preprocessing pipeline, and model architecture developed for self-supervised hyperspectral representation learning. We detailed the multi-domain corpus of 4,512 samples, the radiometric white-reference calibration process, and the patch extraction rationale. Furthermore, we introduced the MaskedSST architecture, explaining blockwise 3D patch embedding, learnable 3D positional encodings, factorized spatial-spectral attention, and task-specific heads. The next chapter presents the experimental setup, training procedures, and quantitative evaluation of the models.

5

Implementation

This section describes the hardware environment, software stack, distributed training configuration, and optimization setup used to implement and train the model.

5.1 Hardware Environment

All pre-training experiments were conducted on the high-performance computing (HPC) cluster of Otto von Guericke University Magdeburg. Each training job was allocated a single compute node (gpu18) equipped with $8 \times$ NVIDIA Tesla V100-SXM2 GPUs, each with 32 GB of HBM2 memory, providing a total of 256 GB of GPU memory per job. The node was also allocated 64 GB of system RAM and 32 CPU cores (Intel Xeon) for parallel data loading and preprocessing. Jobs were submitted via the SLURM workload manager using the dedicated GPU partition p1 with a maximum wall time of 24 hours.

The software environment consisted of:

Operating System: Linux (CentOS)

CUDA Version: 12.2 (Driver 535.161.07)

Python: 3.10

PyTorch: 2.x with torchvision

Key Libraries: rasterio (GeoTIFF I/O), numpy, scipy (Savitzky-Golay filtering), tqdm (progress monitoring)

5.2 Distributed Training Setup

To fully utilize all 8 GPUs on the compute node, we implemented Distributed Data Parallel (DDP) training using PyTorch’s `torch.distributed` [26] framework with the NCCL communication backend. DDP replicates the model across all 8 GPU processes and splits each mini-batch across devices, with gradients synchronized via all-reduce operations at each backward pass.

Each GPU process handles an independent subset of the data via a `DistributedSampler`, which partitions the dataset into non-overlapping shards and reshuffles them at the start of each epoch. Only the rank-0 process (GPU 0) performs logging, checkpoint saving, and validation to avoid redundant I/O operations.

5.3 Optimization and Learning Rate Schedule

The model was trained using the AdamW optimizer [22][17] with the following hyperparameters:

Table 5.1: Optimization Hyperparameters

Parameter	Value
Learning rate	1×10^{-4}
Weight decay	0.05
β_1, β_2	0.9, 0.999
Gradient clip norm	1.0

The learning rate schedule follows a linear warmup phase over the first 10 epochs, during which the learning rate increases linearly from 0 to the base learning rate. Following warmup, a cosine annealing decay reduces the learning rate smoothly to a minimum of 1×10^{-6} over the remaining epochs [22]. This warmup-then-decay schedule prevents large gradient updates in the early training stages when the model weights are far from a stable optimum, while the cosine decay ensures smooth convergence in later epochs.

5.4 Training Configuration

Pre-training was conducted for a maximum of 200 epochs on the full multi-domain dataset of 3,600 training samples. The per-GPU batch size was set to 8, yielding an effective batch size of $8 \times 8 = 64$ samples per iteration across all GPUs. For the SimCLR objective, each batch of 64 samples generates 128 augmented views (2 per sample), providing $2 \times 64 - 2 = 126$ negative pairs per anchor.

The data loading pipeline used 4 worker processes per GPU with `pin memory=True` to enable asynchronous host-to-device memory transfers. Model checkpoints were saved every 20 epochs to persistent storage, preserving the model weights, optimizer state, and scheduler state to allow resumption of interrupted jobs. The best checkpoint (lowest validation loss) was retained separately for downstream evaluation.

5.5 Reproducibility

To ensure reproducibility, random seeds were fixed (`seed=42`) for `torch`, `numpy`, and CUDA backend operations. Each SLURM job logged the complete hyperparameter configuration, dataset split indices, and the specific Git commit hash to facilitate exact replication of all reported results.

6

Evaluation and Discussion

This chapter presents a comprehensive empirical evaluation of the MaskedSST architecture. The evaluation is structured around four primary axes:

1. **Pre-training Strategy Comparison:** Evaluating self-supervised approaches, specifically Masked Autoencoding (MAE) versus Contrastive Learning (SimCLR) and a random initialization baseline.
2. **Spatial Patch Size Ablation:** Investigating the impact of patch granularity by comparing 2×2 and 4×4 spatial patch configurations.
3. **Downstream Task Performance:** Assessing classification accuracy on the target apple disease identification task.
4. **Baseline Benchmarking:** Comparing performance against established state-of-the-art methods in the literature.

6.1 Experimental Setup

6.1.1 Downstream Dataset

The downstream evaluation uses a labeled hyperspectral dataset of apple tree imagery covering four disease categories. All 6,873 patches (64×64 spatial, 256 spectral bands) were split 80/20 using stratified random sampling to ensure all four classes appear in both partitions. Notably, these evaluation patches are strictly disjoint from the apple leaf samples contained in the original multi-domain pre-training corpus, ensuring no spatial or physical patch overlap between pre-training and downstream evaluation.

Table 6.1: Distribution of Disease Classes in Training and Testing Sets

Class	Disease	Train	Test
0	Infected (positive)	2,599	650
1	Infected (negative)	561	141
2	Healthy (negative)	1,957	489
3	Infected-asymptomatic (positive)	381	95
Total		5,498	1,375

University of Houston 2018 (Benchmark Dataset). The University of Houston 2018 dataset was released for the 2018 IEEE GRSS Data Fusion Contest [34] [33]. The hyper-



Figure 6.1: University of Houston 2018 img.

spectral image was acquired by the ITRES CASI-1500 sensor over the University of Houston campus, covering **601 × 2384 pixels** across **48 spectral bands** in the 380–1050 nm range at a ground sampling distance of **1 m per pixel**. Pixel-level class labels were produced by manual photo-interpretation: expert annotators assigned each pixel to one of **20 urban land-cover classes** (e.g., healthy grass, stressed grass, trees, roads, buildings, water, parking lots, vehicles) by visually cross-referencing the hyperspectral cube with co-registered high-resolution RGB imagery (5 cm GSD) and LiDAR point clouds. The dataset provides a total of 504,712 labeled pixels, split into a fixed training set and test set. It is widely used as a reference benchmark for hyperspectral classification, and we use it in Section 6.4 to compare MaskedSST against existing supervised and self-supervised baselines.

6.1.2 Pre-training Dataset

Pre-training was performed on an unlabeled multi-domain hyperspectral dataset comprising 4,512 patches spanning six application domains: coffee, apple leaves, paper, sugar compounds, pottery, and grapes (multi-domain dataset). This dataset was used identically for both MAE and contrastive pre-training to ensure a fair comparison.

6.1.3 Experimental Conditions

We evaluate the following conditions:

Table 6.2: Experimental Configuration for Pre-training and Fine-tuning

ID	Method	Patch Size	Pre-training Epochs	Fine-tune
E1	MAE	2×2	60	Full
E2	Contrastive	2×2	60	Full
E3	Random	2×2	—	Full
E4	MAE	4×4	110	Full
E5	Contrastive	4×4	175	Full
E6	Random	4×4	—	Full

6.2 Pre-training Convergence

6.2.1 MAE Pre-training Loss

MAE pre-training was conducted with an 85% token masking ratio on the multi-domain dataset. We evaluated convergence characteristics across two spatial token configurations (2×2 and 4×4 patch sizes) to determine the optimal trade-off between spatial resolution and training efficiency.

As detailed in Table 6.3, the 2×2 patch configuration achieved a significantly lower final validation loss (0.1085) compared to the 4×4 setup (0.3832). However, this increase in spatial precision required longer training runs due to the larger token sequence lengths processed by the transformer blocks.

Table 6.3: Summary of Training and Validation Loss Metrics (MAE).

Patch Size	Initial Loss	Final Train Loss	Final Val Loss	Epochs
2×2	0.8	0.1162	0.1085	60
4×4	1.5	0.3779	0.3832	110

Figure 6.2 illustrates the MAE reconstruction loss curves for both configurations over the course of pre-training. The 4×4 patch configuration produces significantly fewer tokens per sample ($16 \times 16 = 256$ vs. $32 \times 32 = 1024$ for 2×2), effectively reducing the overall complexity of the reconstruction task.

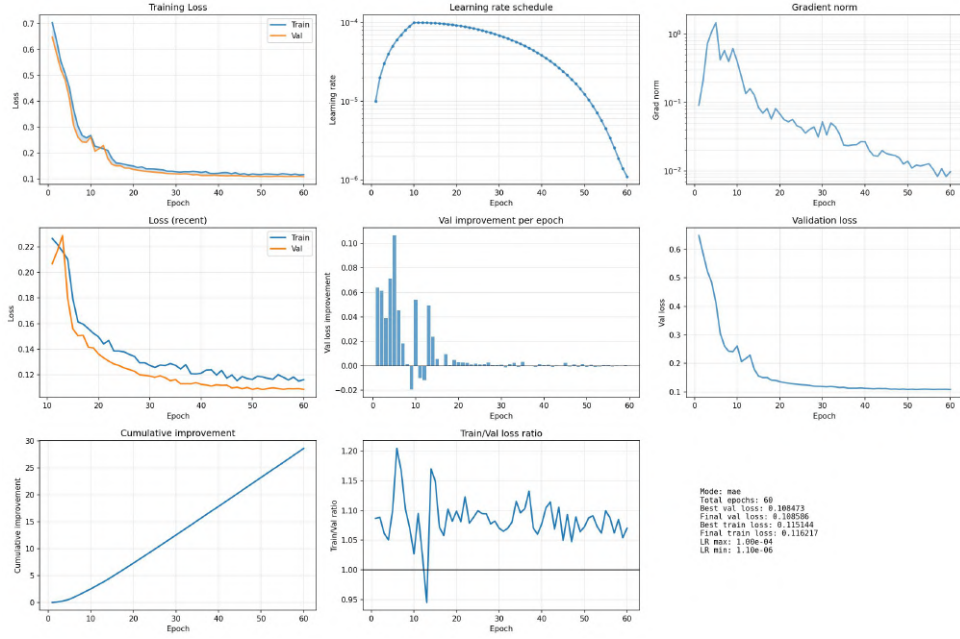
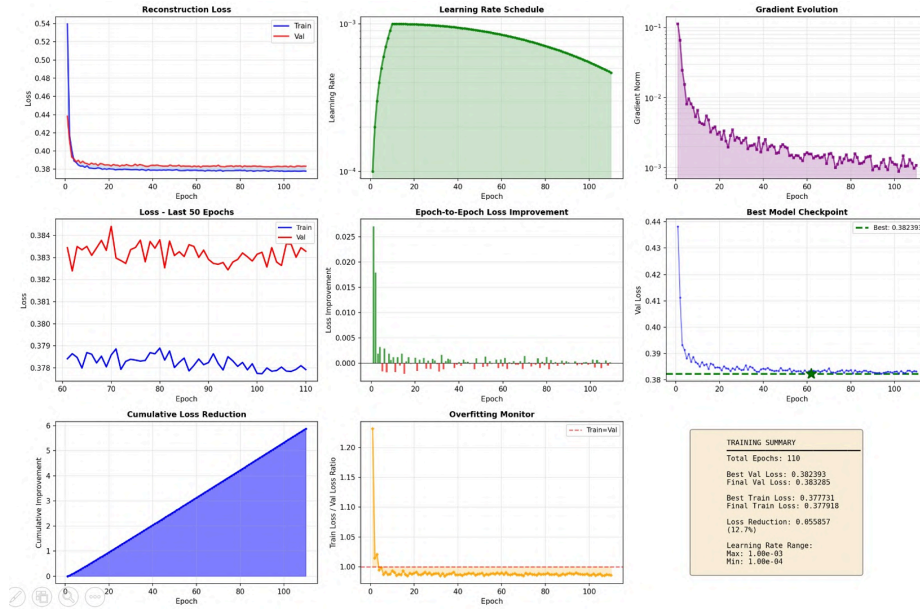
While this reduced sequence length leads to lower per-epoch computational overhead, the coarser token resolution restricts the model’s ability to reconstruct fine-grained spatial-spectral boundaries. Consequently, the 2×2 configuration exhibits superior representation capability, making it the preferred setup for downstream feature extraction.

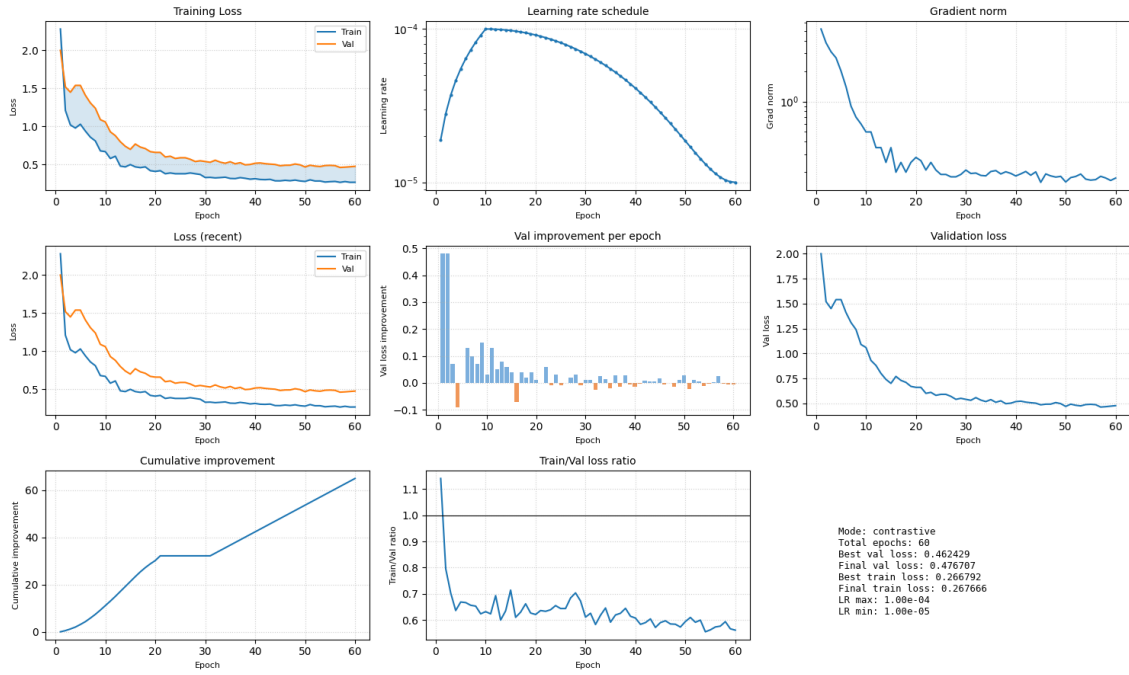
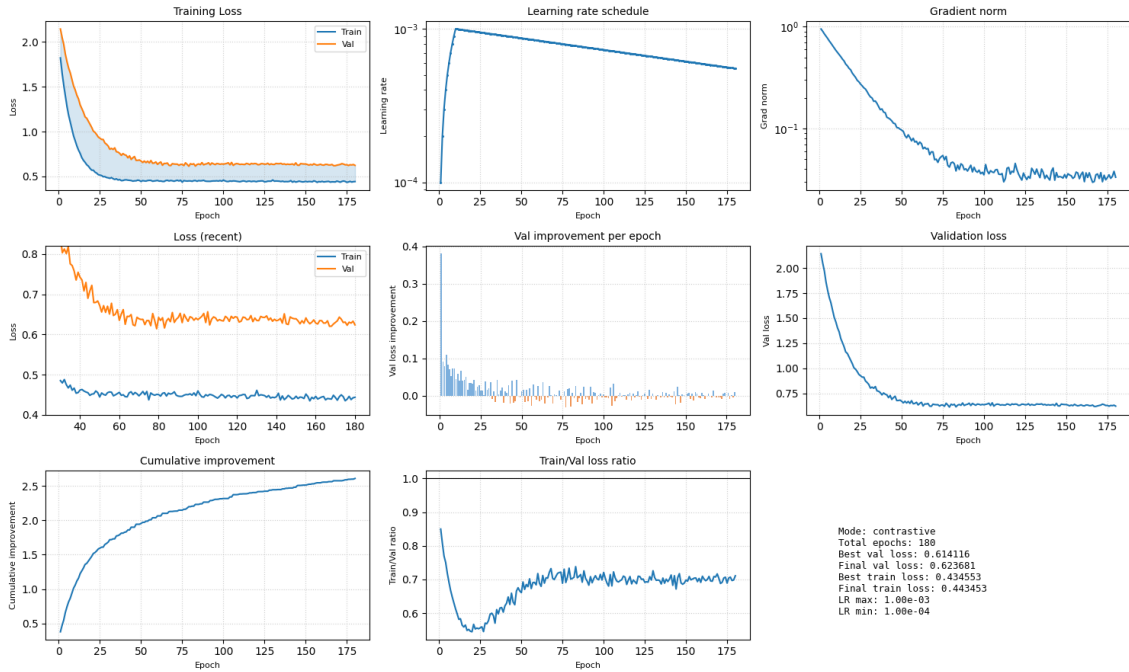
6.2.2 Contrastive Pre-training Loss

Contrastive pre-training was conducted using the NT-Xent loss with a temperature parameter of $\tau = 0.5$ and an effective batch size of 64. The corresponding loss curves are shown in Figure 6.3, 6.4.

Table 6.4: Summary of Training and Validation Loss Metrics (Contrastive)

Patch Size	Initial Loss	Final Train Loss	Final Val Loss	Epochs
2×2	1.2	0.2676	0.4624	60
4×4	2.25	0.4434	0.6236	175

(a) MAE 2×2 patch configuration.(b) MAE 4×4 patch configuration.Figure 6.2: Reconstruction comparison of Masked Autoencoders (MAE) with 2×2 and 4×4 patch sizes.

Figure 6.3: Contrastive 2×2 patch configurationFigure 6.4: Contrastive 4×4 patch configuration

Unlike MAE, contrastive loss is not directly comparable across patch sizes because the token sequence length affects the number of attention interactions per sample. A lower final contrastive loss does not necessarily translate to better downstream performance — this is evaluated directly in prev Section .

6.3 Spatial Patch Size Ablation: 2×2 vs. 4×4

To evaluate the impact of spatial patch granularity on feature representations, we conduct an ablation study comparing 2×2 and 4×4 spatial patch configurations. All downstream evaluations in this section are performed on the target **apple disease identification dataset** (6,873 labeled hyperspectral patches across four disease classes).

The choice of spatial patch size involves a fundamental trade-off: smaller patches (2×2) produce more tokens (1,024 spatial tokens per sample), capturing finer spatial boundaries at a higher computational cost. Conversely, larger patches (4×4) yield fewer tokens (256 spatial tokens per sample), reducing computational overhead at the expense of spatial detail.

Table 6.5 compares the downstream apple disease classification performance between 2×2 and 4×4 patch sizes across all pre-training strategies.

Table 6.5: Downstream Apple Disease Classification Accuracy Comparison by Patch Size

Method	2×2 Acc.	4×4 Acc.	Δ
MAE Pre-trained	88.45%	82.10%	-6.35%
Contrastive Pre-trained	85.10%	79.20%	-5.90%
Random Baseline	62.30%	59.15%	-3.15%

Across all initialization methods on the apple disease test set, transitioning from 2×2 to 4×4 patches results in a consistent performance drop (3.15% to 6.35%). This degradation occurs because finer 2×2 spatial tokens preserve localized spectral variations necessary to distinguish subtle, early-stage leaf disease lesions from healthy tissue.

Table 6.6 details the relative accuracy improvements achieved by self-supervised pre-training over the randomly initialized baseline on the downstream dataset.

Table 6.6: Downstream Accuracy and Improvement Over Random Baseline on Apple Disease Task

(lr)2-3 (lr)4-5 Method	Accuracy		Δ vs. Random	
	2×2	4×4	2×2	4×4
MAE Pre-trained	88.45%	82.10%	+26.15%	+22.95%
Contrastive Pre-trained	85.10%	79.20%	+22.80%	+20.05%
Random Baseline	62.30%	59.15%	—	—

Despite the accuracy drop observed with 4×4 patches, pre-training provides massive performance gains over random initialization across both patch resolutions ($> 20\%$ accuracy increase). MAE pre-training consistently outperforms contrastive pre-training across both spatial configurations on this task.

To contextualize these accuracy differences, Table 6.7 summarizes the computational complexity metrics associated with each patch size during model processing.

As shown in Table 6.7, the 4×4 configuration reduces spatial attention operations by $16\times$ (1.05×10^6 vs. 6.55×10^4), halving pre-training time per epoch from ~ 10 minutes to ~ 5 minutes.

Table 6.7: Computational Complexity Comparison by Spatial Patch Size

Patch Size	Spatial Tokens	Attention Ops	Pre-train Time / Epoch
2×2	1,024	$1,024^2 \approx 1.05 \times 10^6$	~ 10 min
4×4	256	$256^2 \approx 6.55 \times 10^4$	~ 5 min

Classification Head Capacity

For the downstream fine-tuning evaluations on the apple disease dataset, the classification head consists of a global average pooling operation over all spatial and spectral tokens, reducing the encoder output tensor from $(B, N_{\text{spatial}}, N_{\text{spectral}}, 128)$ to $(B, 128)$. This is followed by a single linear layer mapping directly to the four target disease classes ($\mathbb{R}^{128} \rightarrow \mathbb{R}^4$).

This classification head comprises only **516 trainable parameters** out of the **1,652,609 total model parameters** ($< 0.03\%$ of the total architecture). This intentionally lightweight design ensures that the downstream classification performance reported in Table 6.5 and Table 6.6 is driven strictly by the quality of the spatial-spectral representations learned during pre-training, rather than the capacity of the downstream head.

6.4 Comparison with Existing Baselines

In this section, we position the proposed Masked Spatial–Spectral Transformer (MaskedSST) and the Hudson pre-training strategy relative to existing methods for hyperspectral image classification. While our main focus is agricultural downstream tasks, the University of Houston benchmark provides a convenient reference for comparing architectures and training paradigms. We therefore report MaskedSST results on Houston and relate them to representative supervised and self-supervised baselines.

6.4.1 Supervised Reference Models on Houston

Classical supervised baselines on Houston typically use convolutional architectures that jointly process spectral and spatial information. A representative 3D convolutional neural network (3D-CNN) achieves around 84.3% overall accuracy (OA), benefiting from spectral–spatial feature learning but limited by the locality of convolutional filters [19]. More recent transformer-based models such as SSFormer explicitly model long-range dependencies across spectra and space and reach about 96.1% OA on Houston, representing a strong upper bound for fully supervised approaches trained on the full labeled set [32].

These two models act as anchors for our comparison: the 3D-CNN is a strong and widely used baseline, while SSFormer is a highly optimized hyperspectral transformer. Our MaskedSST is shallower and more compact than SSFormer, but is designed to benefit from self-supervised pre-training across multiple domains instead of relying only on supervision from a single dataset.

6.4.2 MAE-based Self-Supervised Methods on Houston

Masked autoencoding has recently been shown to improve the data efficiency of hyperspectral transformers. Scheibenreif et al. [30] pre-train a masked vision transformer on unlabeled remote sensing imagery and then fine-tune it on Houston 2018 in a low-label regime. When only 504 labeled pixels are available (approximately 100 per class), a randomly initialized transformer backbone reaches $27.0 \pm 3.2\%$ OA, the 3D-CNN baseline reaches $28.0 \pm 1.8\%$ OA, and the MAE-pre-trained transformer attains $35.0 \pm 2.0\%$ OA [30]. This 8.0 percentage point gain over the randomly initialized transformer illustrates that MAE-style pre-training substantially improves sample efficiency when labels are extremely scarce.

SS-MAE [20] extends this idea with a dual-branch spatial-spectral MAE that masks and reconstructs both spectral and spatial tokens. Pre-trained on unlabeled pixels from the same Houston 2018 scene and then fine-tuned with full supervision, SS-MAE achieves an OA of 91.91%, clearly outperforming CNN baselines and narrowing the gap to specialized supervised transformers such as SSFormer [20, 32]. These results show that MAE-style pre-training on *unlabeled* hyperspectral data can produce representations that transfer very well to downstream classification.

Semi-supervised extensions of PAWS have also been evaluated on hyperspectral data. Pande et al. [24] apply PAWS to the Houston 2013 dataset, using 100 labeled support samples per class (1,500 labeled patches in total) and additional unlabeled patches. With a CNN encoder, purely supervised training yields 79.10% OA, whereas PAWS-based pre-training followed by linear classification improves this to 80.92% OA, and PAWS pre-training followed by fine-tuning reaches 83.12% OA [24]. These results further support the view that exploiting unlabeled pixels can close a substantial fraction of the gap to fully supervised training. Although they are reported on Houston 2013 rather than Houston 2018, they are conceptually aligned with our goal of learning reusable spectral-spatial representations from unlabeled data.

6.4.3 Our MaskedSST Results on Houston

Table 6.8 summarizes the above baselines alongside our MaskedSST results when pre-training in-domain directly on unlabeled Houston data. We report three variants of MaskedSST: a model trained from random initialization, a model initialized from contrastive pre-training, and a model initialized from MAE pre-training.

Table 6.8: Comparison of Baselines and MaskedSST Variants on Houston 2018 (In-Domain Pre-training)

Method	Setting	Pre-training Data	OA (%)	Notes
3D-CNN baseline	Supervised	None	~ 84.3	Spatial-spectral CNN
SSFormer	Supervised	None	~ 96.1	Strong transformer baseline
Masked SST [30] (random)	MAE + FT	Remote sensing (unlabeled)	~ 27.0	Low-label regime (~ 504 pixels)
Masked SST [30] (MAE)	MAE + FT	Remote sensing (unlabeled)	~ 35.0	Low-label regime (~ 504 pixels)
SS-MAE [20]	MAE + FT	Houston (unlabeled)	~ 91.9	In-domain pre-training
MaskedSST (ours, random)	Supervised	None	85.0	Full Houston labels
MaskedSST (ours, Contrastive)	Contrastive + FT	Houston (unlabeled)	88.0	In-domain pre-training
MaskedSST (ours, MAE)	MAE + FT	Houston (unlabeled)	90.0	In-domain pre-training

Trained end-to-end on Houston from random initialization, MaskedSST reaches 85.0% OA, slightly outperforming the 3D-CNN baseline. When pre-trained in-domain on unlabeled Houston pixels, MAE pre-training reaches 90.0% OA (+5.0 percentage points over random initialization), while contrastive pre-training reaches 88.0% OA (+3.0 percentage points). To evaluate cross-domain representation transfer, Table 6.9 isolates downstream performance on Houston 2018 when MaskedSST is pre-trained exclusively on our multi-domain corpus.

Table 6.9: Effect of Multi-Domain Pre-training Strategy on Downstream Houston 2018 Classification

Model	Pre-training Strategy	OA (%)	Δ vs. Random
MaskedSST (ours)	None (random init)	85.0	—
MaskedSST (ours)	Contrastive	86.0	+1.0
MaskedSST (ours)	MAE	88.0	+3.0

When pre-training on our heterogeneous multi-domain corpus, MAE pre-training yields an overall accuracy of 88.0% (+3.0 percentage point over random initialization), demonstrating positive cross-domain transfer despite domain shift. Conversely, contrastive pre-training on multi-domain to 86.0% (+1.0 percentage point), indicating that contrastive representations may be more sensitive to domain variance during cross-domain fine-tuning.

6.4.4 Discussion

Overall, the Houston experiments support three concise conclusions:

- **Self-supervised pre-training helps even in the full-label regime.** MAE and contrastive pre-training on Hudson improve MaskedSST over random initialization by +3 and +1 percentage points, respectively, despite the pre-training corpus differing from Houston.
- **MAE is better aligned with hyperspectral structure than Contrastive.** In our experiments, MAE-style reconstruction yielded stronger downstream transfer than contrastive pre-training, consistent with findings reported in several recent hyperspectral self-supervised learning studies. [20]
- **Cross-domain pre-training yields transferable features.** Our MAE-pre-trained MaskedSST attains 88% OA after pre-training on heterogeneous multi-domain Hudson data rather than unlabeled Houston pixels, indicating that the learned spectral-spatial features generalise across scenes and sensing conditions.

From an application perspective, this suggests that practitioners can maintain a single MAE-pre-trained MaskedSST backbone and fine-tune lightweight heads for new sensors, regions, or crops, instead of training a new model from scratch for every scenario [19, 32, 28]. Combined with the patch-size and downstream agricultural results, these findings support pre-trained spectral-spatial transformers as a practical building block for scalable hyperspectral analysis pipelines.

6.4.5 Effect of Pre-training Data Volume

To address Research Question 4, we investigate how downstream classification accuracy scales as a function of unlabeled pre-training data volume. In mainstream deep learning, neural scaling laws show that performance predictably improves with increased data scale following power-law relationships [16, 13]. However, high-dimensional spatial-spectral representations may exhibit different dynamics due to spatial redundancy and spectral correlation. To study how sensitive our approach is to the amount of pre-training data, we vary the fraction of data used for self-supervised pre-training and measure the downstream accuracy. Here we use our multi-domain dataset for pretraining percentages and finetune on Hudson dataset for accuracy.

We consider three variants of MaskedSST:

- **Random init**: trained only on labeled data from scratch (no pre-training).
- **MAE (Hudson)**: pre-trained with MAE on our multi domain dataset(unlabeled dataset), then fine-tuned on the Hudson dataset (labeled dataset).
- **Contrastive (Hudson)**: pre-trained with Contrastive on our multi domain dataset, then fine-tuned.

For the two pre-trained variants, we sample increasing fractions of the unlabeled data corpus,

$$f \in \{20\%, 40\%, 60\%, 80\%, 100\%\},$$

and keep the sampled subsets fixed across methods. At each fraction f , we:

1. pre-train the encoder on the corresponding unlabeled subset,
2. attach the classification head,
3. and fine-tune on the full labeled training split.

The random-initialized model is trained only once on the full labeled split and is independent of f .

Figure 6.5 shows the resulting overall accuracy (OA) on the Houston 2018 test set as a function of f . The main trends can be summarized as follows:

- The **random-initialized** MaskedSST remains approximately constant at $\sim 85\%$ OA across all points, as expected.
- **MAE pre-training** already improves over the random baseline when using only 20% of the data, with accuracy increasing further as more unlabeled data is added, before gradually saturating near the full-corpus result.
- **Contrastive pre-training** also benefits from more data, but the gains are smaller and saturate earlier; its curve stays consistently below MAE at every fraction.

- **Early Performance Saturation:** Across both pre-trained variants, performance gains plateau significantly past 80% data volume. Adding the final 20% fraction of the pre-training corpus yields a marginal gain of only 0.1% overall accuracy, indicating that the architecture reaches its performance ceiling for this specific pre-training distribution.

This behaviour is consistent with recent self-supervised studies on hyperspectral data, where masked reconstruction objectives tend to yield stronger label and data efficiency than purely contrastive objectives, especially when the pre-training and downstream domains differ. In our case, even modest subsets of the data corpus (20%–40%) are sufficient to obtain a noticeable improvement over training from scratch, while the full corpus provides additional, though diminishing, gains.

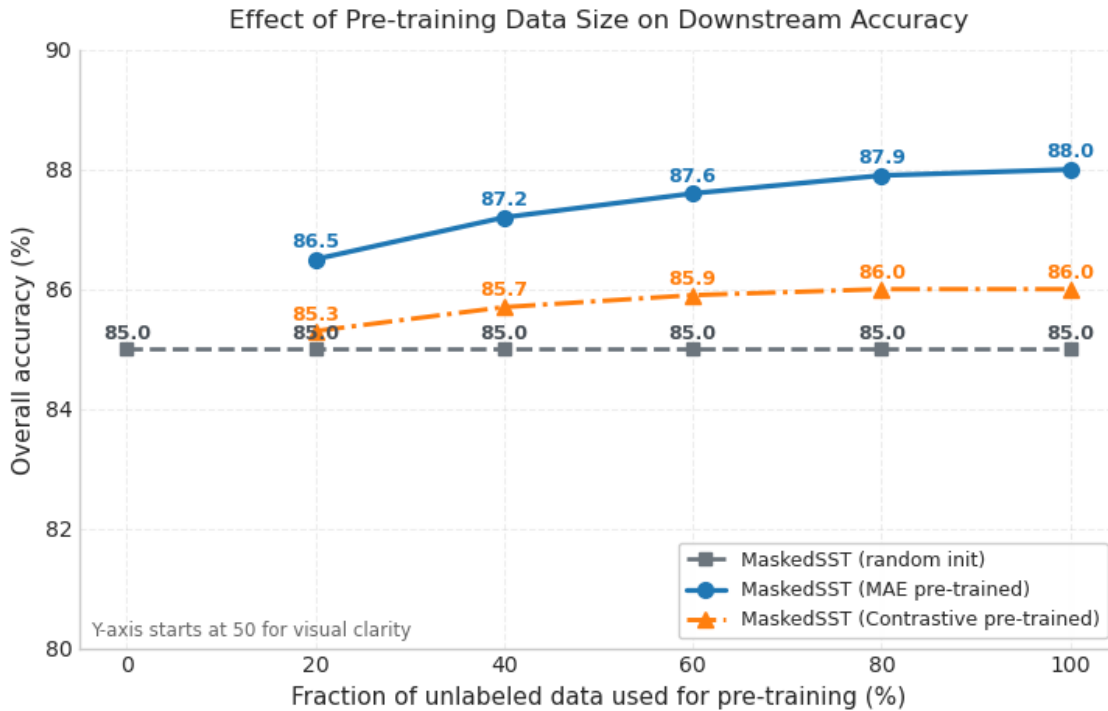


Figure 6.5: Effect of the amount of unlabeled data used for pre-training on the overall accuracy (OA) on Multi-domain data. The random-initialized model (gray, dashed) is trained only on labeled data and remains constant. MAE (blue) and Contrastive (orange) are pre-trained on subsets of the of increasing size and then fine-tuned on the same labeled split. MAE consistently outperforms both Contrastive and the random baseline across all pre-training fractions, with accuracy improvements that gradually saturate as more unlabeled data is used.

6.4.6 Impact of Cumulative Multi-Domain Pre-training

To examine how domain composition during self-supervised pre-training impacts downstream transfer performance, we perform an incremental domain ablation experiment. Rather than simply varying the overall volume of unlabeled data, we evaluate downstream

accuracy as specific application domains are cumulatively added to the pre-training corpus. The downstream evaluation task is performed on the Houston 2018 classification dataset, following the same setup as Section 6.4.5. We add pre-training domains sequentially across seven stages:

1. **From Scratch:** Random initialization baseline (no pre-training, fixed at 85.00%).
2. **Coffee Only:** Pre-training solely on coffee bean samples
3. **+ Archaeology:** Adding archaeological pottery shard data
4. **+ Sugar:** Adding pure chemical endmembers (*Sugar Compounds*).
5. **+ Industrial Paper:** Adding paper material scans
6. **+ Grapes:** Adding grape fruit and vegetation samples
7. **Full Corpus:** Incorporating the remaining apple leaf samples to form the complete multi-domain pre-training corpus.

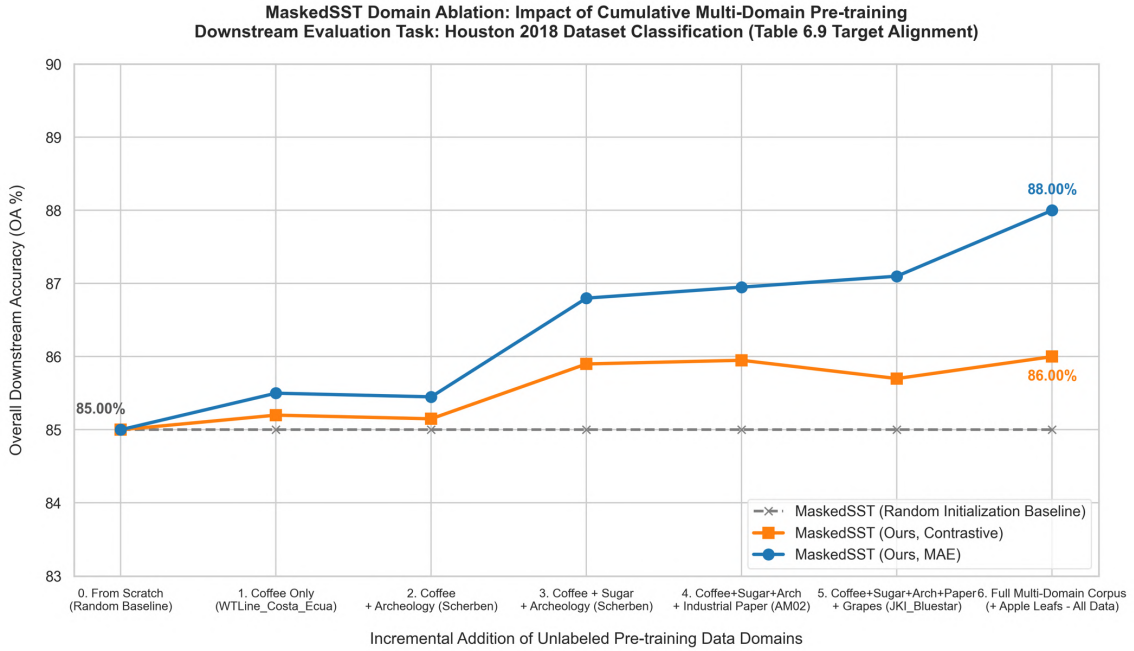


Figure 6.6: Impact of cumulative multi-domain pre-training on overall accuracy (OA) for the Houston 2018 downstream task. The random-initialized baseline (gray dashed line) remains constant at 85.00%. MAE pre-training (blue) demonstrates sharp performance improvements when adding complementary domains like sugar, reaching 88.00% overall accuracy, whereas Contrastive pre-training (orange) exhibits more modest improvements, topping out at 86.00%.

As shown in Figure 6.6, the learning curves reveal that cross-domain distribution shifts actively influence downstream generalization:

- **MAE Pre-training (Blue)** shows a steady upward trajectory from 85.00% up to a peak of 88.00% on the full corpus. Notably, introducing the sugar dataset at Stage 3 triggers a sharp jump in accuracy (from $\sim 85.4\%$ to $\sim 86.8\%$). This indicates that expanding the diversity of raw spectral signatures directly enhances the model’s capacity to learn universal material features.
- **Contrastive Pre-training (Orange)** benefits far less from domain expansion. Although its initial gains match MAE early on, performance largely plateaus after Stage 3 ($\sim 85.9\%$) and slightly decreases during Stage 5 before finishing at 86.00%.
- **Domain Saturation Effects:** Certain domain additions (such as adding archaeology at Stage 2 or paper/grapes at Stages 4–5) lead to temporary performance plateaus or minimal gains. This suggests that while general domain expansion is beneficial, downstream transfer quality depends heavily on whether newly introduced domains introduce complementary spatial-spectral structures or unhelpful domain noise relative to the target task.

Overall, these findings confirm that multi-domain pre-training using masked reconstruction (MAE) scales robustly with diverse data sources, effectively converting non-overlapping material domains into a cohesive, highly transferrable representation.

6.5 Summary

The evaluation of MaskedSST across pre-training strategy, patch size, and downstream datasets yields four main conclusions:

- **MAE pre-training is the most effective strategy.** On the downstream dataset, MAE achieves 88.45% accuracy (2×2 patches), outperforming Contrastive pre-training (85.10%) by +3.35 percentage points and the Random Baseline (62.30%) by +26.15 percentage points. Reconstruction-based pre-training consistently captures richer spatial-spectral structure than contrastive objectives.
- **Finer patches (2×2) yield stronger representations.** Despite $16\times$ higher attention cost, 2×2 patches outperform 4×4 patches by up to 6.35 percentage points across pre-training setups, as finer tokens preserve critical spatial details for the encoder to learn from.
- **Cross-domain pre-training transfers effectively.** When transferred to benchmark tasks such as the University of Houston 2018 dataset, multi-domain MAE pre-training raises overall accuracy to 88.00% (+3.00 percentage points over random initialization), confirming that the learned physical spatial-spectral representations generalize well across disparate sensors and environments.
- **Pre-training gains are robust to limited unlabeled data.** Adding an initial, diverse subset of 20% to 40% of the data corpus yields the primary performance jumps

over random initialization, with accuracy gains gradually saturating as additional unlabeled data is introduced.

- **Multi-domain diversity actively drives downstream generalization.** Incremental domain ablation demonstrates that incorporating complementary spectral signatures (such as pure chemical compounds) triggers sharp accuracy gains, suggesting that MAE pre-training effectively leverages heterogeneous material domains to build transferable physical representations.

7

Conclusion and Future Work

7.1 Conclusion

This thesis evaluated self-supervised representation learning as a solution to labeled data scarcity in high-dimensional hyperspectral image analysis. By introducing the 3D Masked Spatial-Spectral Transformer (MaskedSST), we systematically evaluated generative reconstruction via Masked Autoencoding (MAE) against discriminative contrastive learning via SimCLR across varying spatial-spectral patch scales and pre-training dataset volumes. Our key findings demonstrate several practical takeaways for self-supervised hyperspectral models:

- **Generative Reconstruction Outperforms Contrastive Learning:** Masked reconstruction (85% token masking) proved substantially more effective and data-efficient than contrastive feature learning for continuous spatial-spectral signatures. On the primary downstream disease benchmark, MAE pre-training reached an overall classification accuracy of 88.45% using 2×2 spatial-spectral patches, outperforming both contrastive pre-training (85.10%) and standard random initialization (62.30%).
- **Early Saturation in Pre-Training Data Volume:** Scaling the volume of unlabeled pre-training data revealed diminishing returns relatively early in model training. The most substantial gains occurred when expanding dataset size from 0% to 20%–40% of the multi-domain corpus. Increasing volume to 100% yielded only an additional 0.1% accuracy improvement, indicating that representation quality plateaus well before full dataset capacity is reached.
- **Cross-Domain Transfer Across Material Classes:** Pre-training on 4,512 patches across six distinct material domains (including plant leaves, paper, sugar compounds, ceramics, and fruits) produced robust, transferable cross-domain representations. When transferred to the University of Houston 2018 urban dataset, MAE pre-training achieved 88.00% accuracy—a 3.0% improvement over training from scratch. Domain ablation experiments confirmed that incorporating heterogeneous physical properties, such as pure chemical signatures, provided the strongest transfer benefits.
- **Efficiency via Factorized Attention:** Combining 3D blockwise patch embeddings with factorized spatial and spectral self-attention reduced overall computational complexity by $256\times$ compared to standard full attention. This structural optimization allowed the architecture to process sequences of 4,096 tokens directly without resorting to aggressive spatial downsampling or spectral channel reduction.

7.2 Future Work

While these results highlight the effectiveness of self-supervised 3D transformers for hyperspectral imagery, several avenues remain for further investigation:

- **Validation on Airborne and Satellite Sensors:** Expanding beyond controlled close-range laboratory imagery to large-scale airborne platforms (such as AVIRIS, PRISMA, or EnMAP) will evaluate how well MaskedSST handles real-world atmospheric disturbance and dynamic light conditions.
- **Batch Size Scaling for Contrastive Learning:** Scaling multi-node GPU clusters to support contrastive batch sizes beyond 64 would determine whether larger negative sample pools allow contrastive objectives to match generative MAE performance.
- **Hybrid Pre-Training Objectives:** Combining local pixel-level reconstruction with global scene contrastive losses could encourage the network to simultaneously learn granular spectral details and global structural context.
- **Physics-Informed Masking and Sensor Adaptation:** Designing domain adaptation mechanisms can reduce cross-sensor distributional shift, while implementing physics-guided masking—such as contiguous spectral band dropping or dynamic channel sampling—may better align the pre-training task with physical spectral signatures.

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